NTE: Network Topology Engine

A Hybrid Graph-Relational Engine for Network Topology Modelling

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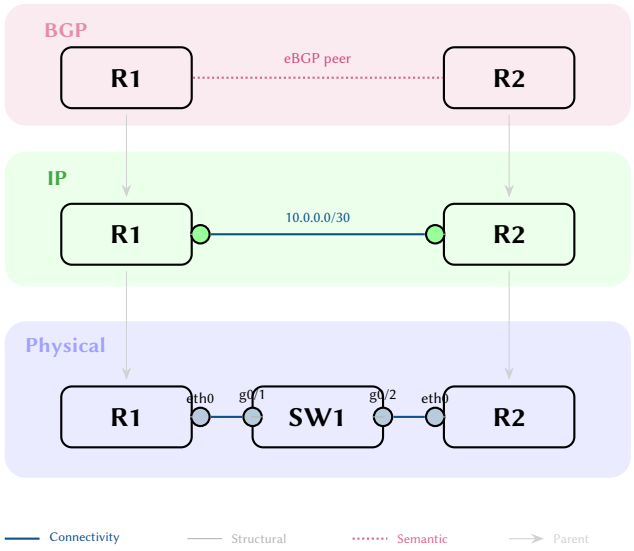


Abstract. NTE (Network Topology Engine) is a hybrid graph-relational engine for hyper-scale structural and property analytics in network topology modelling. Its central design principle is the Dual-Write Consistency paradigm, ensuring that topology structure and columnar properties are mutually isomorphic. This report documents the engine's "Correctness-by-Construction" policy engine, filter-first query planner, and the layered data model that supports configuration synthesis, graph analysis, and live telemetry ingestion.

Contents

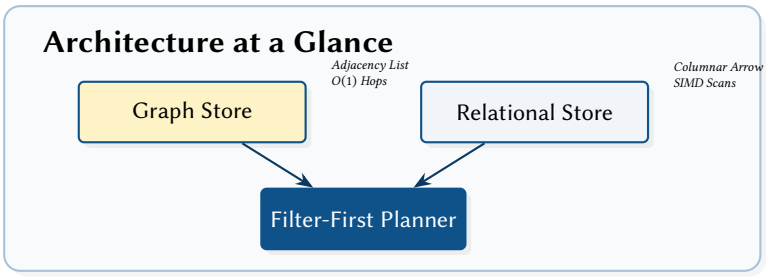
1 Introduction

The scale of modern infrastructure requires a fundamental shift in how topology data is stored and queried.

Insight: *Traditional graph databases excel at structural traversal but fail at columnar analytics; NTE bridges this gap by unifying both representations.*

: Dual-Write Consistency

Every structural mutation to the adjacency graph must be atomically reflected in the columnar property store. This isomorphism ensures that complex queries can be planned using either representation without risk of data divergence.



1.1 What NTE Solves

Network automation tools routinely answer two distinct classes of questions:

1. **Relational / property questions** (e.g., "Which routers have AS 64512 in the SYD PoP?") best answered by scanning structured columns.
2. **Graph / structural questions** (e.g., "Is there a path between A and B under 4 hops?") best answered by traversing adjacency lists.

Most systems optimise for one and bolt on the other. Relational databases support graph queries via recursive CTEs that are hard to optimise. Dedicated graph databases support property lookups but typically store properties row-by-row, lacking columnar performance.

NTE uses a *dual-write* architecture: mutations apply to both a `StableDiGraph` (for structural queries) and Polars Arrow-backed DataFrames (for property scans). A filter-first query planner chooses the optimal representation at runtime.

1.2 Why a Bespoke Engine?

Traditional graph databases (Neo4j, ArangoDB) impose significant overhead for network automation (see Table 1 for a comparative summary):

- **Network overhead**: Millions of small structural validations via wire protocols (e.g., Bolt) introduce unacceptable latency. NTE embeds directly via PyO3, using zero-copy Arrow memory.
- **SIMD Scan Performance**: Most graph DBs store properties row-by-row. NTE's Polars backend matches the SIMD-accelerated columnar scan speeds required for hyperscale analytics.
- **Deterministic Compilation**: The `ank_netcfg` compiler requires a pure-Rust engine to ensure reproducible configuration generation without external daemons.

Table 1. Architectural Positioning: NTE vs. State-of-the-Art.

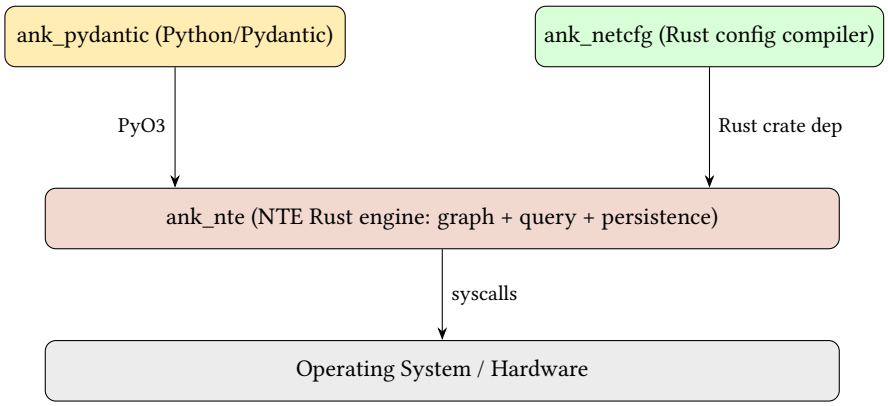
Feature	Neo4j	PostgreSQL	Batfish	NTE
Primary Model	Graph (Row)	Relational (Row)	Simulation	**Hybrid (SIMD/Graph)**
Storage Engine	Native Graph	Heap Tables	In-memory IR	**Arrow/CSR Dual-Write**
Query Primary	Cypher	SQL (CTE)	Declarative	**Filter-First SIMD**
FFI Overhead	High (RPC)	High (RPC)	N/A (Tool)	**Zero (Embedded PyO3)**
"What-If" Speed	Medium (Clone)	Slow (Snapshot)	Fast	**Instant (CoW Shadow)**

1.3 Ecosystem: ank_pydantic and ank_netcfg

NTE is a shared backend for two distinct frontends:

`ank_pydantic` A Python library for interactive exploration and Pydantic-validated modelling.

`ank_netcfg` The **Architectural Synthesis Engine**. A pure-Rust deterministic compiler that Lowering Design Intent into the protocol-specific layered topologies stored in NTE. It handles the "protocol side" of the transformation (e.g., OSPF area allocation, BGP session synthesis).



2 Core Architectural Insights

NTE functions as a specialised "Network Digital Twin Authority," employing several advanced concepts that differentiate its architecture from general-purpose databases.

2.1 Dual-Write Consistency (DWC)

The foundational design constraint of NTE is **Dual-Write Consistency**. To resolve the "pointer-chasing" performance bottleneck common in graph databases during property scans, NTE atomically synchronises every mutation across two distinct in-memory structures:

1. **Structural Store**: A `petgraph::StableDiGraph` that manages topology adjacency and connectivity invariants.
2. **Property Store**: A columnar `DataFrameStore` (Polars) that handles high-dimensional entity attributes (IPs, VLANs, ASNs).

: Atomic Synchronisation

Every topology mutation M must succeed or fail as an atomic unit across both stores. State divergence is prevented by a `DualWriteGuard` that rolls back the structural graph if the columnar property update fails.

This hybrid approach allows NTE to leverage SIMD-accelerated relational scans for candidate pruning (Phase 1) while maintaining O(1) neighbor access for structural traversals (Phase 2).

2.2 Multi-Source Topology Aggregation

The fundamental architectural principle of NTE is **multi-source aggregation** (Figure 1). Topology intent from heterogeneous sources—design diagrams, IP address management spreadsheets, configuration files, live telemetry feeds—is normalised into a single, canonical **Blueprint** that serves as the engine's source of structural truth.

: Blueprint Isomorphism

The Blueprint acts as the single source of absolute structural truth. The mapping from various input sources into this formal model is a deterministic aggregation. The engine then compiles this singular blueprint, fanning it out into the multi-layered topology overlays (L1, L2, L3) that make up the internal database.

This principle decouples *input format* from *structural representation*. Once the blueprint is materialised, the NTE engine maps it to the layered topology by applying Design Rules (e.g., automatically generating the required interfaces, loopbacks, and cabling). The same blueprint can equally drive configuration synthesis, graph analysis, or live telemetry correlation.

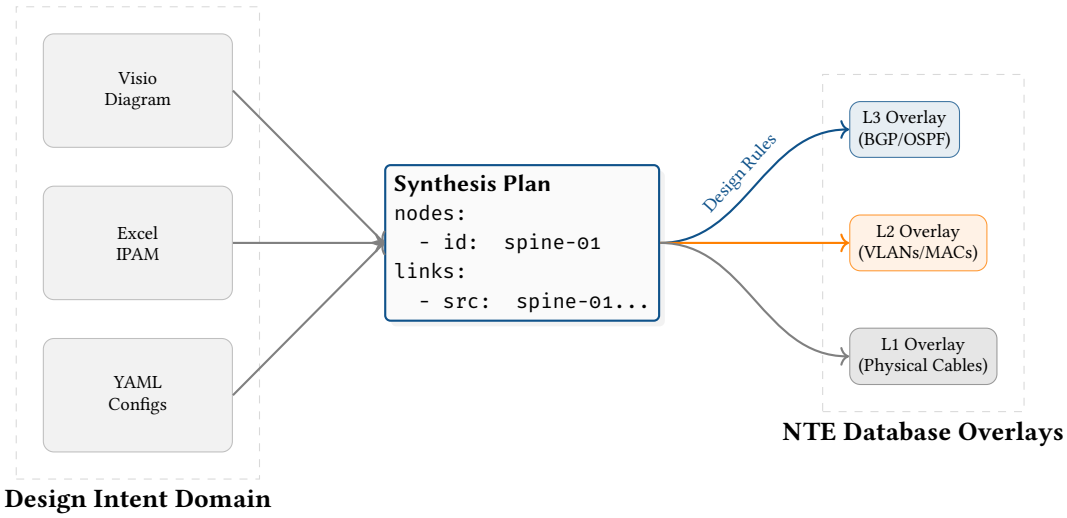


Figure 1. Multi-source aggregation: diverse input sources are normalised into a singular formal Blueprint, which the engine compiles and fans out into rigorous, multi-layered topology overlays.

2.3 Bimodal Execution: Strict vs. Relaxed

While NTE can act as a deterministic compilation engine for configuration synthesis (via `ank_netcfg`), restricting the system exclusively to rigid, top-down generation would significantly limit its utility. Modern network operations increasingly demand ad-hoc, real-time analysis of active network states (e.g., ingesting streaming BGP telemetry, or analysing live routing tables).

To support both design-time generation and run-time analysis, NTE implements a **Bimodal Execution Architecture** (Figure 2):

Strict Mode (The Design Authority) In this mode, NTE enforces the full blueprint aggregation model. All structural mutations are subjected to pre-commit schema validation and invariant checks. If a user attempts to draw a logical edge between two nodes that lack underlying physical `Connectivity` connectivity, the transaction is forcefully aborted. This mode is used when NTE acts as the source-of-truth compiler generating configuration artefacts.

Relaxed Mode (The Telemetry Sink) In this mode, structural invariants and schema requirements are dynamically bypassed. Users can rapidly stream ad-hoc properties (e.g., live SNMP counters or streaming telemetry) directly into the Polars DataFrames, or draw arbitrary graph edges representing discovered control-plane states, without satisfying the strict underlying Endpoints Model requirements. This transforms NTE into a high-performance graph-analytical sink for live operational data.

This bimodal capability ensures that the exact same query engine (NTE-QL) and visualisation tooling can be used to analyse both the "intended" network (Strict) and the "actual" network (Relaxed) using identical semantics.

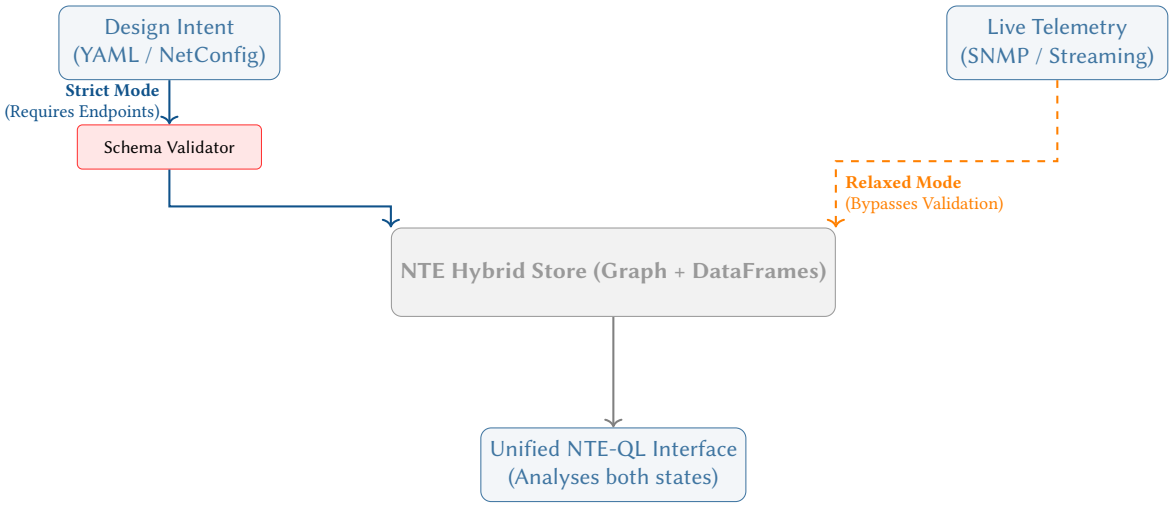


Figure 2. Bimodal Execution Architecture. Strict mode forces design intent through schema validation (ensuring the Endpoints model is perfectly adhered to). Relaxed mode bypasses validation, allowing live telemetry to stream directly into the engine. Both modes share the same unified NTE-QL interface.

2.4 Flexible Ingestion Patterns

To accommodate both top-down design and bottom-up discovery, NTE supports two primary data ingestion patterns. These patterns correspond to the **Bimodal Execution** modes described in Section 2.3.

2.4.1 Pattern A: The Explicit Endpoints Model

Used primarily in **Strict Mode**, this pattern requires explicit modelling of nodes, endpoints, and structural edges. It is used when the topology must serve as a high-fidelity digital twin for physical failure simulation. The ingestion sequence follows the structural hierarchy: `add_nodes` → `add_endpoints` → `add_structural_edges` → `add_connectivity_edges`.

2.4.2 Pattern B: Ad-Hoc Direct Links

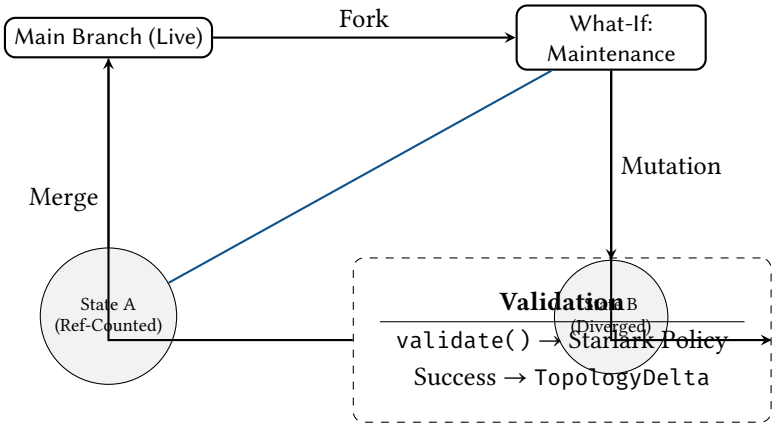
Used in **Relaxed Mode**, this pattern allows drawing direct semantic edges between device nodes via the `add_direct_links` method. This bypasses the need for explicit endpoint modelling, enabling the rapid ingestion of discovered control-plane state (e.g., from an ARP table or a BGP peer list). This pattern is also suitable for modelling simple internal sub-components (like a Line Card connected directly to a Chassis) where the full complexity of physical interface pinning is not required.

While this pattern has lower structural fidelity, it remains fully compatible with the NTE-QL query engine. The Query Planner automatically selects the most efficient traversal path: if a direct edge exists, it is matched; otherwise, the planner attempts to expand the query into a 3-hop endpoint traversal.

2.5 "What-If" Branching (Git for Networks)

Standard network management systems manipulate a single, mutable "current state." Simulating a failure or testing a new configuration typically requires spinning up a separate, heavyweight staging environment or running offline simulations.

Because NTE's storage layer is built on a hybrid architecture of `petgraph` and `Polars` DataFrames, it benefits inherently from **Copy-on-Write (CoW)** semantics.



When a user requests a "What-If" branch (e.g., to simulate a maintenance window):

1. **Memory Efficiency**: The underlying Arrow memory buffers backing the Polars DataFrames are not duplicated; they are reference-counted.
2. **Isolated Mutations**: As the user drops nodes (simulating a failure) or adds edges (simulating a new link) in the branch, only the modified data structures are copied and diverged. The `SnapshotManager` handles these isolated instances.
3. **Verification**: The Starlark policy engine (`validate()`) can be run against this isolated branch. If the simulation results in a Single Point of Failure (SPOF), the maintenance plan can be programmatically rejected.
4. **Patch Generation**: If the branch is successful, the `diff_against` engine computes the `TopologyDelta` between the branch and the live state, generating a precise "Change Request" payload.

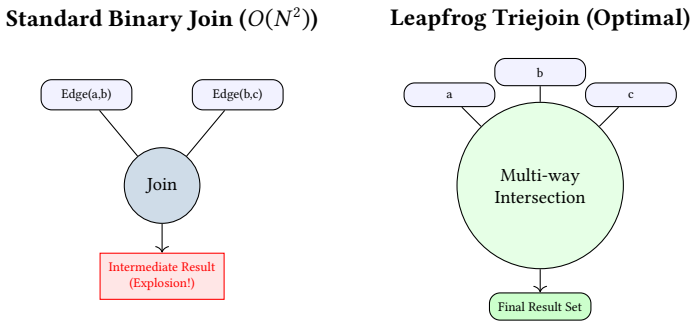
Case Study: Peak Traffic Resilience A Tier-1 ISP needs to upgrade the firmware on a core backbone router during a 4-hour maintenance window. Using NTE, they fork the live topology into a "What-If" branch and programmatically `drop_node()` the router. The Starlark policy engine instantly detects that the resulting topology violates a `min_path_redundancy(3)` constraint for high-priority financial traffic. The engineer adjusts the maintenance window to a time when secondary links can absorb the load, avoiding a potential multi-million dollar SLA breach.

2.6 Worst-Case Optimal Joins (WCOJ)

Traditional relational and graph databases execute queries using binary joins (joining two tables/edge sets at a time). While optimal for trees or linear paths, binary joins suffer catastrophic intermediate result explosion when querying highly cyclic structures (like triangles or cliques).

Modern network fabrics (e.g., Leaf-Spine, BGP Route-Reflector meshes, optical rings) are fundamentally cyclic. A query to find a redundant triangle `MATCH (a)->(b)->(c)->(a)` in a dense fabric using binary joins can take $O(N^2)$ time, even if the final result set is extremely small.

NTE's query engine is planned to incorporate **Worst-Case Optimal Joins (WCOJ)**, specifically the Leapfrog Triejoin algorithm.

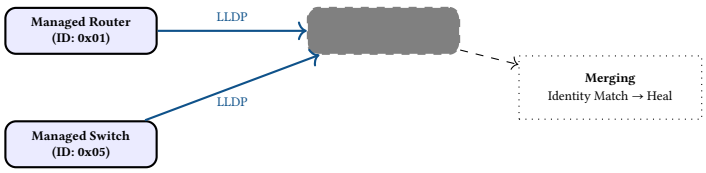


Rather than joining edges pairwise, WCOJ joins all variables simultaneously using multi-way intersection over sorted iterators (enabled by our `CSR` cache and `FixedBitSet` structures). This guarantees that the execution time is bounded by the *theoretical maximum size of the output*, completely eliminating the intermediate result explosion.

Case Study: Hyperscale Fabric Verification In a 10,000-node Leaf-Spine data center, an operator runs a query to identify all potential 3-cycle feedback loops in BGP peering. A standard graph database generates millions of intermediate pairs before filtering for the third edge, causing high CPU usage and multi-second latency. NTE's WCOJ engine performs a simultaneous intersection of the edge iterators, returning the 45 actual loops in under 50ms, enabling real-time drift detection during automated configuration rollouts.

2.7 Shadow Graph Inference (The "Known Unknowns")

A persistent challenge in network digital twins is the "edge of the map." Telemetry data (LLDP, CDP, BGP peering logs) often reveals links to devices that are not managed by the central orchestrator. NTE embraces the reality of incomplete data through **Shadow Graph Inference**.

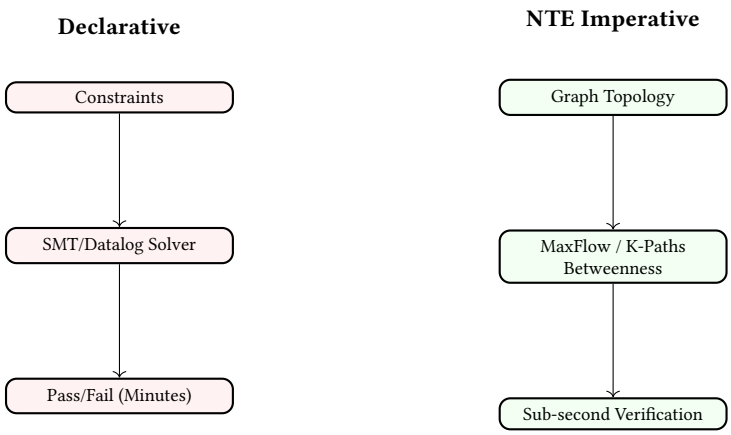


When an edge points to an unknown ID, the engine dynamically spawns a "Shadow Node" tagged with a `confidence_score`. These translucent nodes visualise the boundary of managed territory. As more telemetry arrives, or when the device is officially provisioned, the "Ship of Theseus" semantic identity engine automatically heals the Shadow Node into a fully managed Node, migrating all associated edges without disruption.

Case Study: Mapping the Invisible An enterprise network relies on a third-party managed optical transport network (OTN) for inter-datacenter connectivity. While the enterprise has no management access to the OTN switches, their edge routers report LLDP neighbors on the optical hand-offs. NTE creates "Shadow Nodes" for these OTN devices, allowing the enterprise's digital twin to visualise the full end-to-end path. When a fiber cut occurs, NTE's path analysis correctly identifies the "Invisible" transport hop as the common point of failure, even though it wasn't in the official inventory.

2.8 Differential Reachability: Imperative vs. Declarative Approaches

A core goal of network verification is proving that reachability policies hold before and after a change. While tools like Batfish translate configurations into logical constraints (Datalog/SMT), NTE takes an imperative algorithmic approach.



NTE leverages extreme-performance graph algorithms written in Rust:

K-Shortest Paths (Yen's Algorithm) Calculates the top K paths. A behavioral diff reveals shifts in path length and redundancy.

Max Flow / Min Cut (Dinic's Algorithm) Instantly calculates theoretical maximum edge-disjoint capacity between points.

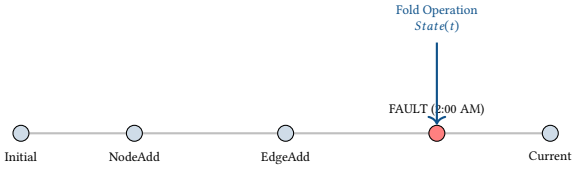
Betweenness Centrality Identifies topological bottlenecks dynamically.

By exposing these algorithms to the Starlark policy engine, NTE provides "sub-second" structural verification, enabling real-time feedback loops impossible with heavy logic solvers.

Case Study: Real-Time Compliance A healthcare network's compliance policy requires that the "Medical Imaging VLAN" must have at least two physically diverse paths across the campus backbone. During a planned fiber maintenance event, an engineer uses the NTE UI to simulate a link shutdown. The imperative engine calculates the Max Flow between the imaging centers in 12ms and warns the engineer that diversity has dropped to zero. This real-time feedback loop allows the engineer to reroute traffic *before* the fiber is cut, ensuring HIPAA compliance.

2.9 The "Forensic Time Machine" (Event-Sourced Temporal Queries)

The Write-Ahead Log (WAL) primarily built for durability implicitly provides an event-sourced architecture enabling true "Time-Travel" debugging. Because every mutation is a discrete `TopologyEvent`, the state at any point in history is a mathematical fold: $\text{State}(t) = \text{InitialState} + \sum \text{Events}[0..t]$.

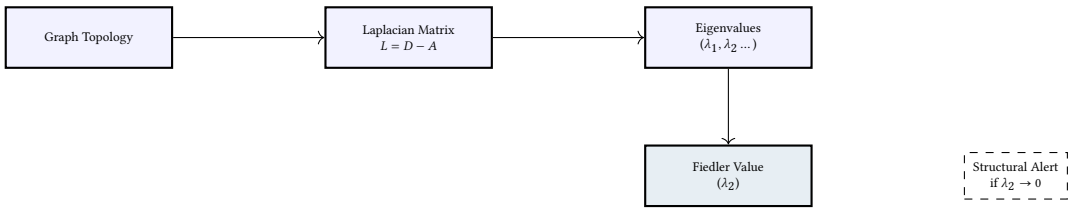


Instead of saving thousands of discrete snapshots, NTE instantly reconstructs point-in-time topologies. Engineers can run standard queries against the exact topology the network experienced during a failure, fundamentally changing root-cause analysis.

Case Study: The 2:00 AM Ghost A financial services provider experiences intermittent loss of connectivity at 2:15 AM on a Sunday. By 8:00 AM, self-healing protocols have masked the problem. Using the NTE Forensic Time Machine, the team enters the timestamp. The engine reconstructs the state, revealing a brief fiber maintenance that triggered a BGP misconfiguration existing only in that specific topological permutation.

2.10 Spectral Graph Signatures (Non-AI Anomaly Detection)

While Graph Neural Networks (GNNs) are powerful, they suffer from "black box" explainability issues. NTE utilizes **Spectral Graph Theory** to extract mathematical signatures of topological health by analyzing the eigenvalues of the Graph Laplacian matrix ($L = D - A$).



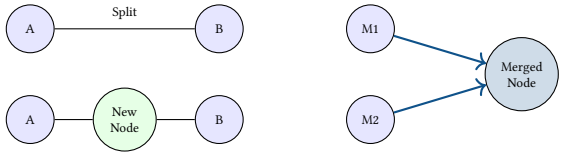
The second smallest eigenvalue (λ_2), the **Fiedler Value**, summarises how well-connected the network is. If a configuration diff cause the Fiedler value to drop by 40%, the engine instantly flags a massive structural fragmentation risk. The corresponding Fiedler Vector identifies specific "choke point" links.

Case Study: Early Warning of Network Fragmentation A SCADA network for a national power grid remains operational during cascading failures due to redundancy. However, NTE detects that the Fiedler Value has dropped from 0.45 to 0.02, indicating the network is "algebraically brittle." NTE alerts operators to a critical loss of structural integrity hours before the final redundant link would have failed.

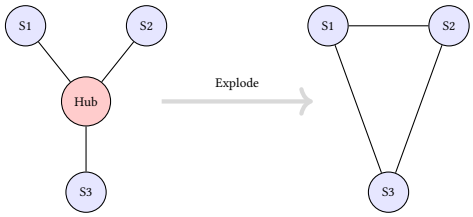
2.11 Structural Transformations: Split, Merge, Explode, and Aggregate

NTE provides high-level **Structural Transformations** that allow complex topological manipulation with single API calls. These are atomic mutators implemented in Rust and Python.

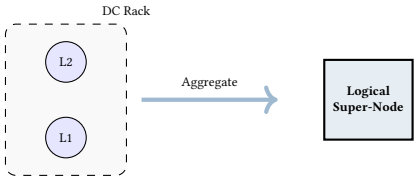
Split and Merge Split inserts a new node into an edge, handling endpoint creation and rewiring. Merge collapses two nodes into one, inheriting all connections.



Explode (The Hub-and-Spoke Dissolver) Explode removes a central hub and replaces it with a full mesh between neighbors, invaluable for simulating bridge removal while maintaining logical connectivity.

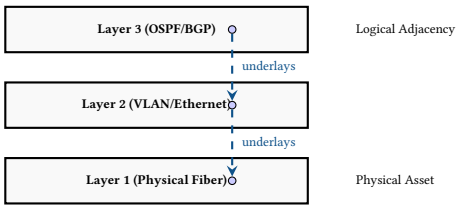


Aggregate (Graph Summarization) Aggregation algorithmically collapses a cluster of nodes (e.g., an OSPF Area) into a single logical Super-Node.



2.12 Cross-Layer Lineage: Vertical Impact Analysis

A defining feature of a digital twin is mapping dependencies between layers. NTE uses **Vertical Lineage Links** to connect endpoints across the Physical, Data-Link, and Network layers.

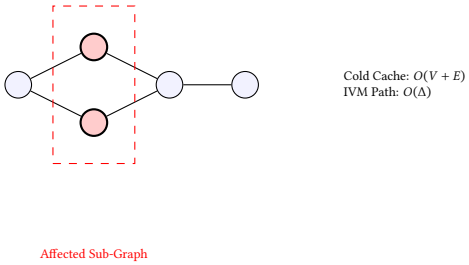


Case Study: Root-Cause of a "Silent Drop" An ISP experiences intermittent packet loss. Vertical lineage reveals an MTU mismatch on a physical fiber member of an LACP bundle. Querying `MATCH`

`(b:BGP)-[:UNDERLAYS*]->(p:Physical) WHERE p.mtu != b.expected_mtu` instantly identifies the port causing silent drops for jumbo frames.

2.13 Incremental View Maintenance (IVM): Computational Efficiency

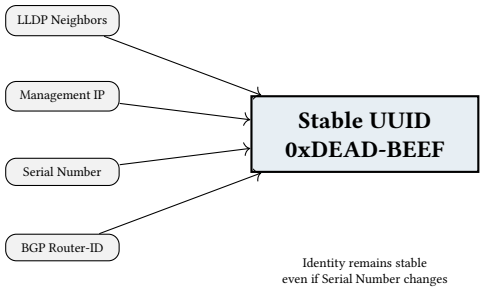
Recalculating global properties for massive graphs is expensive. NTE employs **Incremental View Maintenance (IVM)** to update views in sub-millisecond time by identifying the "Impacted Sub-Graph" via the WAL.



Case Study: Real-Time Policy Monitoring A global financial network requires order execution traffic to have 40Gbps Max-Flow capacity. NTE's IVM engine detects local flaps and recalculates local Max-Flow in 0.8ms, maintaining a continuous compliance dashboard in real-time.

2.14 The "Ship of Theseus" Semantic Identity Engine

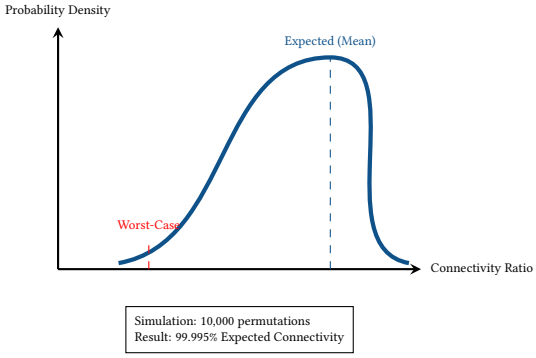
NTE separates **Semantic Identity** from volatile attributes (IP, MAC). An "Identity Resolver" uses multi-source telemetry to maintain stability.



Case Study: Zero-Downtime Inventory Migration A DC hardware refresh replaces all Spine switches. Because NTE identifies Spines semantically, historical data and branches map to new devices automatically, preserving lineage for trend analysis across generations.

2.15 Probabilistic Risk Modeling: Monte Carlo Analysis

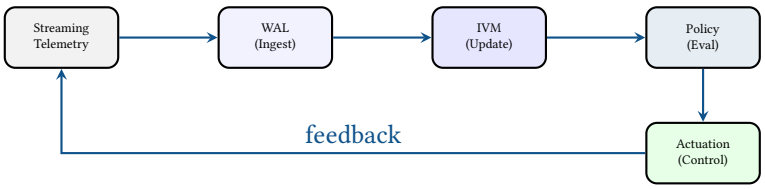
NTE integrates a high-performance **Monte Carlo Engine** to quantify topological risk across stochastic multi-link failure permutations, assigning a Reliability Score to the topology.



Case Study: Calculating "Blast Radius" in SD-WAN Before deploying routing policies for 5,000 SD-WAN sites, NTE simulates 50,000 failure events. The resulting distribution reveals a hidden blast radius where a specific regional failure would isolate 15% of stores, allowing proactive redundant link provisioning.

2.16 Reactive Policy Loops: Streaming Graph Ingestion

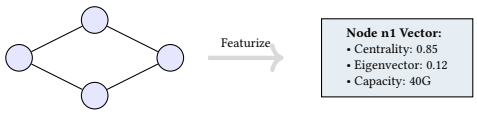
NTE supports **Reactive Control Loops** that ingest telemetry streams and actuate changes. Events enter the WAL, trigger IVM updates, and are evaluated by the Starlark policy engine.



Case Study: Sub-Second "Algebraic" Traffic Shedding NTE monitors Spectral Signatures (λ_2). A link flap drop in λ_2 triggers a policy that instantly sheds non-critical bulk services, preventing cascades in under 100ms.

2.17 Graph-Native Featurization: Enabling ML-Aided Networking

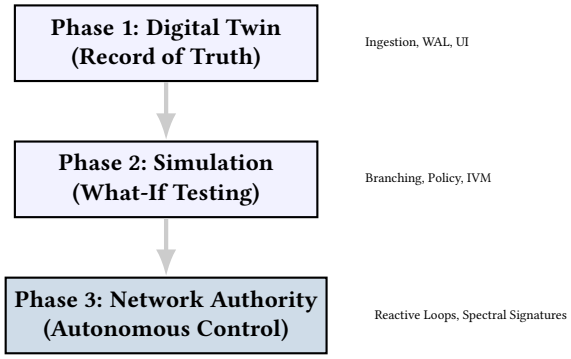
NTE exports **Graph-Native Featurization** vectors for nodes, containing metrics like Betweenness Centrality, Spectral Eigenvectors, and Max-Flow, acting as a feature store for Graph Neural Networks (GNNs).



Case Study: Predictive Congestion in 5G Slices An operator feeds NTE featurized fronthaul topology into a GNN to predict congestion hotspots 10 minutes in advance with 94% accuracy, enabling dynamic slice scaling.

2.18 Strategic Roadmap: The "Autonomous Network" Vision

NTE aims for a transition from a passive Digital Twin to an **Autonomous Network Authority**.



§§3–4 cover the core data model and implementation architecture. §6 describes the storage engine and §7 provides an algorithm reference for the key graph and relational primitives. §8 covers the query system: the filter-first planner, the fluent Python query builder, NTE-QL, and the interactive REPL. §9 covers ACID transactions and conflict detection. The remaining sections — persistence, GPU acceleration, graph algorithm plugins, reactive events, production diagnostics, and Raft clustering — are advanced deployment topics that can be read independently and in any order.

3 Data Model

Source

Axiom 1 (Dual-Write Atomicity). *Every mutation to the topology must be reflected atomically in both the graph adjacency store and the columnar property store. A transaction is only committed if both representations are successfully updated.*

Theorem 1 (Graph-Relational Equivalence). *The set of all nodes V and edges E reachable via graph traversal is equivalent to the set of rows in the primary Polars DataFrames. $\forall v \in DiGraph, \exists! row \in DataFrame$ mapping to v.*

3.1 Formal Definition

A topology in NTE is a typed, directed, attributed multi-graph $G = (V, E, \ell_V, \ell_E, \pi_V, \pi_E, L, \lambda)$ where:

$$
\begin{aligned}
V &= V_{\text{node}} \cup V_{\text{endpoint}} \quad \text{(disjoint node and endpoint sets)} \\
E &\subseteq V \times V \quad \text{(directed edges)} \\
\ell_V &: V \to \Sigma_V \quad \text{(type label, e.g. “Router”, “Switch”)} \\
\ell_E &: E \to \{\text{Connectivity}, \text{Structural}, \text{Intranode}, \ldots\} \quad \text{(edge kind)} \\
\pi_V &: V \to (\text{String} \rightharpoonup \text{JSON}) \quad \text{(property maps)} \\
\pi_E &: E \to (\text{String} \rightharpoonup \text{JSON}) \quad \text{(edge property maps)} \\
L &\quad \text{(ordered set of named layers)} \\
\lambda &: V \to L \cup \{\bot\} \quad \text{(optional layer assignment)}
\end{aligned}
$$

Nodes are assigned integer IDs that are globally unique within a topology instance. NTE uses `i32` as the external ID type. IDs are chosen by the caller; NTE does not auto-assign them (though helpers exist for auto-increment).

3.2 Vertex Types

NTE recognises two top-level vertex categories and one specialisation.

> **Terminology note**
>
> NTE uses “node” specifically for *network devices* and “endpoint” for their connection points. This differs from graph-theory usage where “node” means any vertex. In this document, “vertex” is used when the graph-theoretic meaning is intended.

3.2.1 Nodes

A **Node** represents a network *device* — a router, switch, server, firewall, optical transponder, etc. (Figure 3). Nodes carry a string `node_type` and an optional layer membership. Property data is stored as an arbitrary JSON object, populated by `ank_pydantic` from Pydantic model fields.

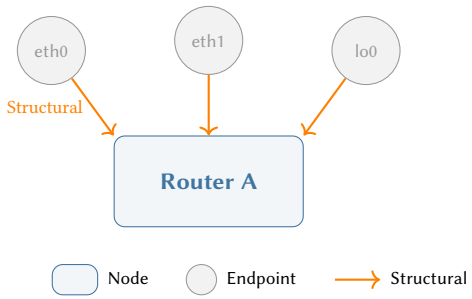


Figure 3. A Node with three Endpoints. Each **Structural** edge points *from* the Endpoint *to* its owning Node, expressing the ownership relationship.

3.2.2 Endpoints

An **Endpoint** is a connection point on a node (Figure 4). Endpoints fall into three broad categories:

Physical endpoints represent hardware attachment points where a cable terminates: Ethernet ports, fibre connectors, transceiver slots. They correspond to tangible components on a device.

Logical endpoints represent protocol-level connection points that exist in software rather than hardware: loopback interfaces, VLAN sub-interfaces, tunnel endpoints (e.g. GRE, VXLAN VTEPs). A single physical port may host multiple logical endpoints via sub-interfacing.

Generic endpoints are used when the physical/logical distinction is irrelevant or unknown — for example, a telemetry source reporting an interface name without classifying it. Generic endpoints allow ingestion pipelines to populate the topology without forcing premature classification.

Each endpoint carries an optional `endpoint_name` (e.g. `GigabitEthernet0/1`, `lo0`, `Vlan100`) for human identification, alongside a type label (e.g. `"Endpoint"`, `"Interface"`) and an arbitrary property map.

> **: Endpoint Ownership Invariant**
>
> Every endpoint is owned by exactly one node via a Structural edge. An endpoint cannot exist unattached. This invariant is enforced at creation time and preserved by all mutation operations.

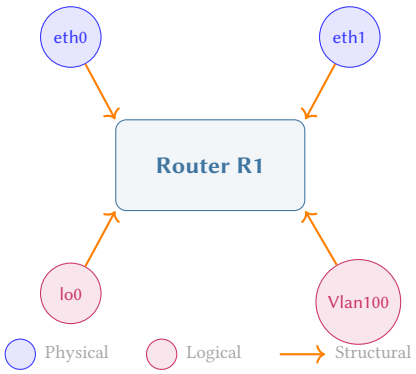


Figure 4. A device with mixed physical and logical endpoints. Physical endpoints (`eth0`, `eth1`) represent hardware ports; logical endpoints (`lo0`, `Vlan100`) represent software-defined connection points. All are owned by the same node via Structural edges.

3.2.3 Logical Nodes

A **LogicalNode** (Figure 5) is a specialised node that *always* belongs to a layer and represents protocol-level or virtual topology objects that do not correspond to physical hardware. Typical examples include BGP session state machines, OSPF areas, VLAN broadcast domains, and VPN routing/forwarding (VRF) instances.

Key distinction from regular Nodes. A LogicalNode *requires* a layer assignment (non-optional), whereas a regular Node may exist without one. This mandatory binding ensures that every logical object is anchored in a specific plane of abstraction, enabling the engine to reason about cross-layer dependencies.

Query visibility. LogicalNodes are excluded from default query results to avoid polluting device-centric queries. Targeting them requires explicitly matching their layer — for example, `MATCH (s:BGPSession) WHERE s.layer = "bgp"` in NTE-QL. This opt-in visibility prevents protocol-state objects from appearing in infrastructure queries that expect only physical devices.

Common patterns. A LogicalNode in the BGP layer linked via a `Parent` edge to a physical Router represents "this router participates in this BGP session." Similarly, an OSPF area LogicalNode may have Parent edges to all routers in that area, providing a single query target for area-wide analysis.

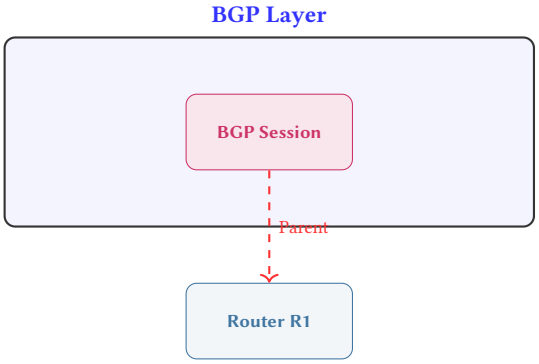


Figure 5. A LogicalNode (BGP Session) residing within a layer plane, linked to its parent physical Node via a **Parent** edge. Logical nodes are excluded from default queries unless their layer is explicitly targeted.

3.3 Edge Kinds

NTE defines three principal external edge kinds. Internally, the engine also uses several bookkeeping edge kinds (`Parent`, `Child`, `Root`, `Origin`) that are invisible to Python callers by default.

3.3.1 Structural (Ownership)

A **Structural** edge (Figure 6) runs *from* an endpoint *to* the node that owns it. It expresses the "this port belongs to this device" relationship. A node may own any number of endpoints; each endpoint has exactly one owning node. The Python method is `add_structural_edges`.

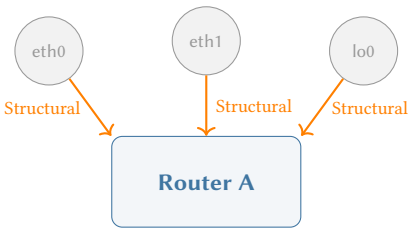


Figure 6. Structural (Ownership) edges. Three Endpoints are owned by a single Node. Arrows point *from* Endpoint *to* Node, expressing the "belongs to" relationship.

3.3.2 Connectivity

A **Connectivity** edge (Figure 7) connects two endpoints. It models a physical or logical link between two ports on different devices. Connectivity edges are bidirectional by convention: NTE stores them as a pair of directed edges. The Python method is `add_connectivity_edges`.

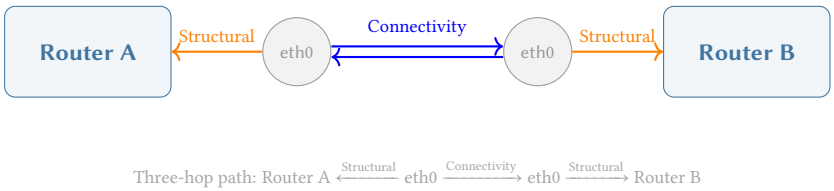


Figure 7. Connectivity edges between two devices. Each device connects via its Endpoint. A bidirectional link is stored as two explicit directed edges. The three-hop traversal path is shown below.

3.3.3 Intranode

An **Intranode** edge connects two nodes within the same logical chassis or container. Where Structural edges express *ownership* (an endpoint belongs to a device), Intranode edges express *co-location* (two nodes share a physical chassis or logical container). These are fundamentally different relationships: ownership is hierarchical and one-to-many; co-location is peer-to-peer and many-to-many.

Use cases.

- **Multi-VRF routers**: each VRF is a separate Node, but all inhabit the same physical chassis. Intranode edges link VRF nodes to the Global Routing Table node (see Figure 8).
- **Modular chassis**: line cards are child Nodes connected to the chassis supervisory module via Intranode edges.
- **Virtual router instances**: multiple VR instances inside a hypervisor, sharing the same compute host but appearing as independent nodes.

Direction convention. Intranode edges are directed from child to parent (e.g. VRF → Global RT), following the same "points toward the owner" convention as Structural edges. Bidirectional peer relationships (e.g. VRF A ↔ VRF B for route leaking) are modelled as two explicit directed edges.

> **Implementation Note**
>
> The current Rust implementation unifies Intranode under the `Structural` edge kind in petgraph, with the semantic distinction tracked via event types (`IntraEdgeCreated` vs `IntranodeEdgeCreated`). This pragmatic simplification avoids a separate edge-kind enum variant whilst preserving the conceptual separation at the API layer. The Python method is `add_intranode_edges`.

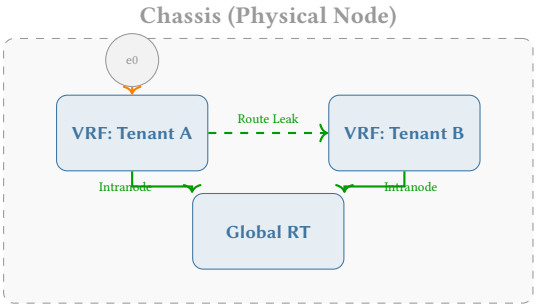


Figure 8. Intranode edges within a chassis. VRF sub-nodes connect to the Global Routing Table via **Intranode** edges. Each sub-node can optionally own its own Endpoints (shown on VRF: Tenant A). The dashed edge represents a route-leak relationship.

3.3.4 Device Shortcut (DeviceLink)

When a Connectivity edge is created between two endpoints, NTE automatically adds a **DeviceLink** shortcut edge between the owning devices. This avoids the three-hop traversal (*DeviceA* → *PortA* → *PortB* → *DeviceB*) in the common case where callers only want device-to-device connectivity. The shortcut is maintained in `device_shortcuts: HashMap<(i32,i32), Vec<EdgeIndex>>` inside `NteGraph`.

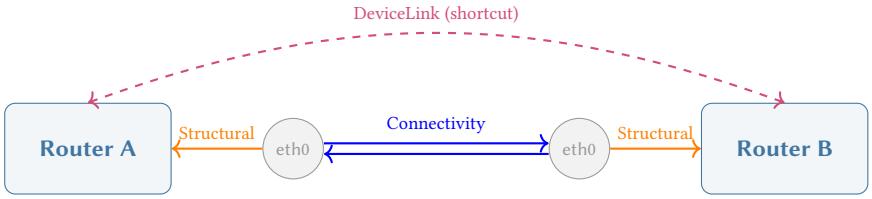


Figure 9. The DeviceLink shortcut. The dashed purple edge bypasses the three-hop Structural–Connectivity–Structural path, providing direct device-to-device adjacency for common queries.

3.3.5 Semantic (Named Virtual Edges)

A **Semantic** edge carries a named label (e.g. `Semantic("bgp")`, `Semantic("ospf_area_0")`) and represents a user-defined logical relationship that exists purely at a named layer. Unlike Connectivity edges, which connect endpoints, Semantic edges connect *nodes directly* and bypass the Endpoints model entirely.

Use case. Semantic edges overlay protocol relationships that do not correspond to physical cables. For example, an OSPF adjacency between two routers that share a broadcast domain, or a management-plane "monitors" relationship between a collector and its targets. In each case, the relationship is meaningful at the logical layer but has no single physical attachment point worth modelling.

Trade-off. Semantic edges sacrifice the blast-radius precision of the Endpoints model for query ergonomics. Because they skip the three-hop pattern, failure propagation through Semantic edges cannot isolate individual port-level failures. They should be used when the physical attachment point is irrelevant to the analysis.

Python API. The method `add_semantic_edges(layer, source_ids, target_ids)` creates Semantic edges within the specified layer. The `layer` parameter must name an existing registered layer.

3.4 Composite Patterns

The vertex types and edge kinds defined above combine into recurring structural patterns that appear throughout real network topologies.

3.4.1 The Three-Hop Pattern

The most fundamental traversal in NTE is the *three-hop pattern*: to walk from one device to another, the path always passes through Device A $\xleftarrow{\text{Structural}}$ Endpoint $\xrightarrow{\text{Connectivity}}$ Endpoint $\xrightarrow{\text{Structural}}$ Device B. This pattern is a direct consequence of the Endpoint model and is the reason DeviceLink shortcuts exist (see Figure 9).

> **Abstraction Leak**
>
> The necessity of the three-hop pattern highlights an abstraction leak: the physical reality of network ports complicates basic queries. This is precisely why NTE-QL introduces **Virtual Cypher Translation**

(Section 8.3), which automatically collapses these endpoint traversals when a user executes queries at the device level, removing the cognitive burden from the engineer.

3.4.2 Endpoint Lifecycle

In practice, endpoints are created and wired through a predictable sequence:

1. **Creation**: Endpoints are typically created alongside their owning node in a single batch operation (e.g. `add_nodes` with an `endpoints` parameter). The engine atomically creates the node, its endpoints, and the Structural ownership edges in one transaction.
2. **Wiring**: Connectivity edges are added between endpoints on different nodes to establish links (`connect_endpoints`).
3. **DeviceLink shortcut**: NTE automatically creates the DeviceLink shortcut edge when a Connectivity edge is added, requiring no explicit action from the caller.
4. **Validation**: In Strict mode, endpoint creation requires specifying the parent node and layer. The engine validates ownership consistency and layer compatibility before committing.

3.4.3 Internal Structure (Chassis, VRFs, Line Cards)

Nodes can optionally expose internal sub-structure. A router may be treated as an atomic device, or it may be decomposed into child nodes representing line cards, forwarding engines, or VRF instances. This decomposition is modelled as child Nodes connected via Intranode edges within a chassis boundary (see Figure 8).

The internal structure is entirely optional: a node with no Intranode edges behaves as an opaque, atomic device. When sub-structure *is* present, each child node can own its own Endpoints, enabling per-line-card or per-VRF connectivity modelling.

Because **Intranode** edges represent structural ownership rather than external connectivity, they are implemented in the Rust backend as a subset of the `Structural` edge kind. In NTE-QL, users can explicitly query these internal sub-structures by matching against the `:Structural` relationship:

```
// Find all VRFs contained within a specific physical chassis
MATCH (chassis:Router)-[:Structural]->(vrf:VRF)
WHERE chassis.hostname = "core-1"
RETURN vrf.name, vrf.route_target
```

Because this query specifically requests a `Structural` edge rather than a semantic relationship type (like `:eBGP`), the Query Planner recognises this as an explicit internal query and **bypasses Virtual Cypher expansion**. This allows operators to seamlessly dive into the sub-components of a node without switching query languages.

3.4.4 Cross-Layer Relationships

A topological digital twin must track how a high-level logical construct (like a BGP session) relies on lower-level physical infrastructure (like a fibre cable). NTE achieves this via **Parent** edges, which span across the independent Layer planes.

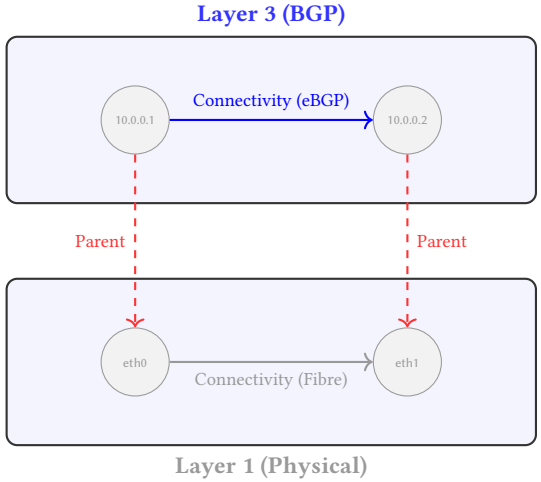


Figure 10. Cross-Layer Relationships. L3 BGP Endpoints are bound to L1 Physical Endpoints via **Parent** edges. If `eth0` fails, the engine traverses the incoming Parent edge to deterministically tear down the dependent BGP session.

3.4.5 Failure Domain Isolation (Blast Radius)

The primary justification for the strict verbosity of the Endpoints Model is deterministic failure propagation. When a physical or logical component fails, the engine must accurately calculate the "blast radius"—the exact subset of downstream services affected.

By modelling ownership explicitly via `Structural` edges, failure simulation becomes a pure mathematical reachability problem (Figure 11).

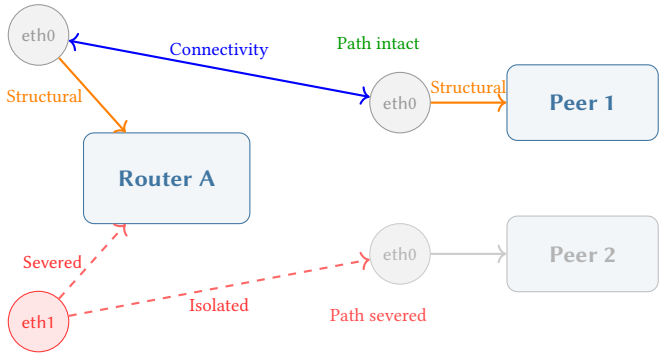


Figure 11. Blast Radius Propagation. When `Line Card 2` fails, its outgoing `Structural` edge is severed. The connection to Peer 2 is isolated whilst Peer 1 remains unaffected because `eth0` and `eth1` reside on different endpoints with intact Structural edges. A naive `(Router)-[CONNECTED_TO]->(Router)` model cannot express this partial failure.

3.5 Layer Hierarchy

Layers partition the topology into independent sub-graph planes (Figure 12). Each layer defines its own set of nodes and endpoints that form an independent Connectivity graph. Common examples include a `physical` layer (hardware links), an `ip` layer (IP adjacencies), a `bgp` layer (BGP sessions), and an `mpls` layer. Layers can share the same physical nodes — a Router may appear in both the Physical and BGP layers — but each layer's edges are independent.

Layer dependency DAG. Layers declare parent-child dependencies (e.g. BGP depends on IP, IP depends on Physical). NTE validates at registration time that these dependencies form a directed acyclic graph; cycles are rejected with a `LayerCycleError`. This DAG drives the cross-layer traversal primitives (`get_ancestor_in_layer`, `get_descendant_in_layer`) described in Section 3.6.

Interaction with the Endpoints model. Within a single layer, the standard three-hop pattern applies: Device $\xleftarrow{\text{Structural}}$ Endpoint $\xrightarrow{\text{Connectivity}}$ Endpoint $\xrightarrow{\text{Structural}}$ Device. *Across* layers, Parent edges connect a higher-layer endpoint or node to its lower-layer counterpart. For example, a BGP session endpoint at L3 has a Parent edge pointing to the physical Ethernet endpoint at L1 that carries it (see Figure 10).

Strict vs Relaxed enforcement. In **Strict** mode, `connect_endpoints` validates that both endpoints belong to the same layer. In **Relaxed** mode, this constraint is lifted to support ad-hoc telemetry ingestion where layer metadata may be incomplete or inconsistent. The mode is configured per-topology at creation time.

Base layer. A special `base` layer acts as a cross-cutting plane for nodes that must appear in all layers — for example, a management collector that aggregates data across every protocol plane. Nodes assigned to the base layer are implicitly visible in every other layer's query scope.

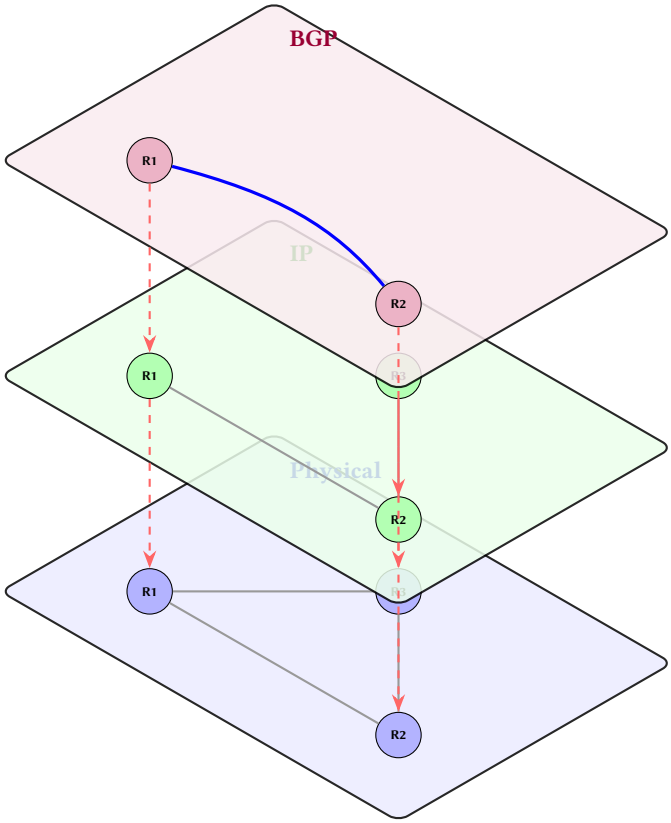


Figure 12. The 3D layered topology model. Each plane uses a distinct colour (blue = Physical, green = IP, purple = BGP). Node labels (R1, R2, R3) are visible inside each circle. Red dashed arrows show cross-layer structural dependencies projecting logical overlays down onto their physical underlays.

Layers are stored internally in a `LayerDependencyGraph` (a DAG of layer names). At edge-creation time, NTE validates that nodes on incompatible layers are not mixed (raising `LayerMismatchError` if so). The topological ordering of layers also determines the default iteration order for cross-layer algorithms.

3.6 Multi-Layer Topology Invariants

A fundamental strength of the NTE engine is its ability to maintain structural invariants across the vertical stack. Most network models treat physical and logical layers as isolated graphs; NTE formalises their relationship through **Ancestor/Descendant** mappings.

: Cross-Layer Structural Integrity

Every logical node n_L in layer L_i must have a structural mapping $\mathcal{M} : n_L \to n_P$ where n_P is an underlay node in layer $L_j (j < i)$. If a physical node is removed, all its logical descendants must be either re-parented or marked as "orphaned" to maintain referential integrity.

NTE provides optimised primitives for "tunnelling" through the stack:

- `get_ancestor_in_layer(node_id, layer)`: Efficiently walks the hierarchy to find the physical hardware (e.g., Switch Port) supporting a high-level session (e.g., BGP Peer).
- `get_descendant_in_layer(node_id, layer)`: Projects a physical failure up the stack to identify all impacted logical services.

This cross-layer awareness ensures that queries like *"Find the physical switch port supporting this BGP session"* are resolved in $O(H)$ time, where H is the height of the layer stack, typically ≤ 5.

3.7 State Reconciliation: Intent vs. Observed

A critical requirement for modern network automation is the ability to distinguish between the **Intent** (what the designer wants) and the **Observed State** (what the telemetry reports). NTE treats these as two distinct but isomorphic representations of the same topology.

: Truth Isomorphism

Every node and edge in NTE carries a `source` attribute, either `intent` or `observed`. The engine provides built-in `reconcile()` primitives that compute the graph-structural and property-relational "Drift" between these two worlds.

By storing both Intent and Observed state in the same hybrid graph-relational engine, NTE enables sophisticated drift analysis:

- **Structural Drift**: Detecting missing links or unexpected hardware that exists in the physical world but not in the blueprint.
- **Property Drift**: Identifying configuration mismatches where the operational speed or MTU of an interface disagrees with the design.

The result of a reconciliation operation is a `TopologyDelta` that can be used to either "push" the intent to the network or "pull" the observed state back into the blueprint to reflect reality.

3.8 Property Maps

Both nodes and edges carry arbitrary key-value properties stored as JSON objects. Properties are serialised from Pydantic model instances by `ank_pydantic` and stored in Polars DataFrames as individual columns (one column per field name). The Polars representation enables fast columnar scans, type-aware comparisons, and SIMD acceleration.

Warning

Property columns are created lazily as new field names appear. This means queries against fields that have never been set on any node return an empty result rather than an error. Check column existence before relying on absent-field semantics.

4 Implementation Architecture

NTE enforces a strict layered crate architecture to prevent cyclic dependencies and to separate the graph connectivity fabric from business-logic concerns. Pure domain types sit at the bottom of the stack; the event-sourcing primitives that drive incremental view maintenance are pushed down to the foundation layer so that the query engine and topology manager can consume them independently. This layering is the structural prerequisite for the reactive pipeline described in later sections.

4.1 Workspace Layout

NTE is an 18-crate Cargo workspace. The core crates and their roles are detailed in Appendix A (Table 5).

Figure 13 shows how the crates relate.

4.2 Crate Dependency Hierarchy

To maintain high architectural integrity and support reactive IVM, NTE utilises a strict layered dependency model. Following the Phase 20 refactor, all shared mutational types (Events and Sequences) were migrated to `nte-graph` to eliminate cyclic dependencies between the query engine and the business logic layer.

1. **Layer 0 (Domain)**: `nte-domain` contains pure data structures with zero external dependencies.
2. **Layer 1 (Foundation)**: `nte-graph` provides the fundamental connectivity fabric, memory layouts (CSR), and the event sourcing definitions.
3. **Layer 2 (Execution)**: `nte-query` implements the relational and graph execution engines (WCOJ, IVM) on top of foundation types.
4. **Layer 3 (Business Logic)**: `nte-topology` provides high-level topology management, transactions, and formal verification.
5. **Layer 4 (Bindings)**: The `src/` root crate provides the PyO3 interface to Python.

The policy engine (Section 5) operates at Layer 3, leveraging the query engine at Layer 2 to evaluate design-rule contracts against the topology.

5 Policy Engine

5.1 Design Rationale

Network topologies must satisfy design-rule constraints: every edge device must have at least two uplinks, no two devices in the same rack should be connected to the same spine, all transit links must have ≥ 100 Gbps speed. Encoding these constraints as ad-hoc Python checks is fragile and scattered across the codebase.

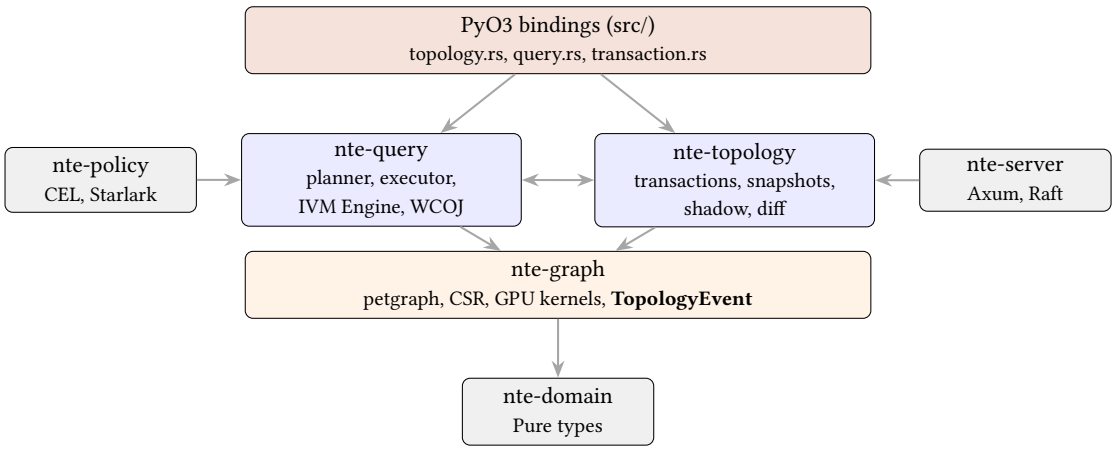


Figure 13. The Layered Crate Architecture. Shared foundation types (Events) are pushed down to `nte-graph` to enable clean, uni-directional dependency flow between the query engine and topology manager.

NTE externalises design-rule contracts into a declarative policy layer [1] (Figure 15), separating *what the constraint is* from *when it is evaluated*. Policies can be evaluated on demand, before a mutation (what-if), or as a continuous validation step in a CI pipeline.

: Shadow Topology Isolation

NTE leverages its copy-on-write (CoW) architecture to support "What-If" analysis via Shadow Topologies. A mutation can be applied to an ephemeral child snapshot, validated against the full policy engine, and discarded without ever modifying the primary production state. This provides Correctness-by-Construction for network-wide changes.

5.2 Policy Specification

While NTE supports multiple formats for policy definition, the preferred representation is NTE-QL due to its declarative clarity and structural expressiveness. A policy is essentially a named query that identifies "violations" in the topology.

For example, a policy ensuring all devices meet a 100G speed requirement can be expressed as a query for non-compliant nodes:

Listing 1. Policy expressed as an NTE-QL violation query

```
// P001: All devices must support 100G
MATCH (n:Device)
WHERE n.speed_gbps < 100
RETURN n.id, n.speed_gbps
```

Structural invariants, such as ensuring high-availability for external connectivity, are similarly expressed:

Listing 2. Structural policy: transit exit redundancy

```
// P002: Every PoP must have at least 2 transit exits
MATCH (p:PoP)
WHERE count { (p)-[:HAS_EXIT]->(e:TransitExit) } < 2
RETURN p.id, count { (p)-[:HAS_EXIT]->(e:TransitExit) }
```

When integrated into a CI/CD pipeline, these queries are executed by the `PolicyEngine`. Any non-empty result set triggers a policy violation finding. For legacy integration and configuration management, a YAML-based specification is also supported (see Appendix B).

`attribute` Evaluated per node/edge. The expression is checked against each entity's property map independently.

`structural` Evaluated once against global topology metrics (counts, ratios, aggregate properties).

5.3 The Starlark Policy Engine

For policies requiring programmatic logic beyond simple comparisons, NTE includes a Starlark-based policy engine. Starlark is a Python-like scripting language designed for safe, deterministic execution in embedded environments. It was chosen because:

- **Sandboxed**: no filesystem access, no network access, no mutable global state.
- **Deterministic**: no random number generation, no timing functions.
- **Familiar**: syntax is a subset of Python, comfortable for network engineers.
- **Fast**: scripts are parsed and compiled to bytecode once, then executed against each context.

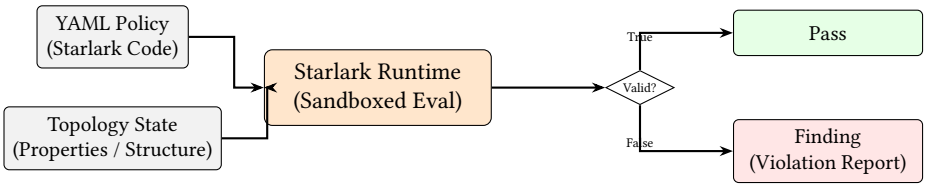


Figure 14. Starlark Policy Evaluation. The sandboxed runtime safely evaluates Python-like policy code against the topology state to generate deterministic compliance findings.

Listing 3. Starlark policy example: uplink redundancy

```
# Policy: every leaf switch must have at least 2 uplinks
def validate_leaf_uplinks(node_id, uplink_count, required):
    if uplink_count < required:
        fail("Node {} has only {} uplinks, need {}".format(
            node_id, uplink_count, required
        ))
    return True

# Host function bound from Rust: check_uplinks(node_id, required)
# Returns True if the node has >= required uplinks
result = check_uplinks("leaf-01", 2)
result  # Starlark scripts return the value of their last expression
```

5.4 Policy Evaluation (CEL & Starlark)

The `PolicyEngine` struct pre-compiles all policy expressions at load time. To balance performance and expressivity, NTE employs a tiered execution model:

CEL (Common Expression Language) Used for high-performance attribute and structural checks. CEL is a C++-like declarative language designed for safe, fast evaluation. It is ideal for simple property comparisons and existence checks.

Starlark Used for complex procedural logic that requires loops, conditionals, or custom functions (e.g., recursive path validation).

At evaluation time, the engine performs the following steps:

1. **Global Context Injection**: Structural policies are evaluated once against a global `RuntimeContext` containing aggregate topology metrics (e.g., total node count, layer distribution).
2. **Subject Iteration**: Attribute policies are evaluated for every matching subject (node or edge). The engine binds `this` (the entity ID) and `attrs` (the property map) into the evaluation scope.
3. **Deterministic Reporting**: Each violation produces a stable `Finding` report. These reports are deterministic and can be exported as JSON (v1 schema) for automated reconciliation.

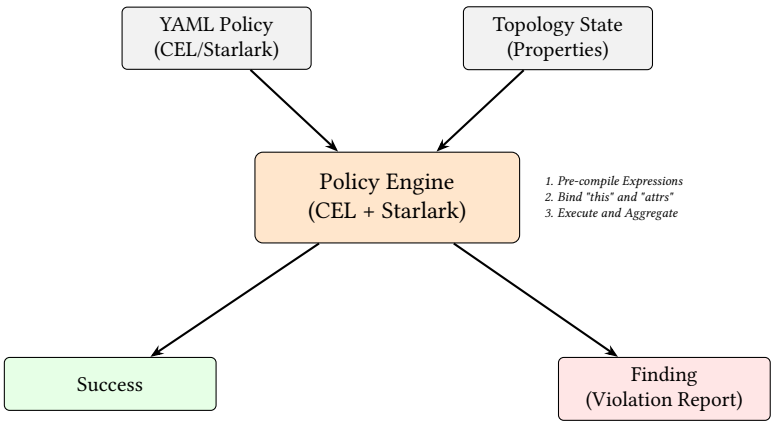


Figure 15. Policy Evaluation Pipeline. The engine pre-compiles declarative CEL expressions and procedural Starlark logic, executing them against a unified topology context to generate structured findings.

5.5 Shadow Topologies and DeltaNet Validation

NTE supports *what-if* analysis via `ShadowTopology`: a clone of the live topology onto which a proposed `TopologyDelta` is applied without affecting the real state.

Warning

The Snapshot Memory Tax: While the Polars DataFrames utilise Arrow CoW buffers (making property cloning effectively $O(1)$), `petgraph::StableDiGraph` is backed by standard Rust `Vec`s. Creating a shadow topology requires an $O(|V|+|E|)$ deep memory copy of the entire structural graph. For topologies exceeding 100,000 nodes, creating a shadow (or starting a transaction) will incur a millisecond-level CPU stall.

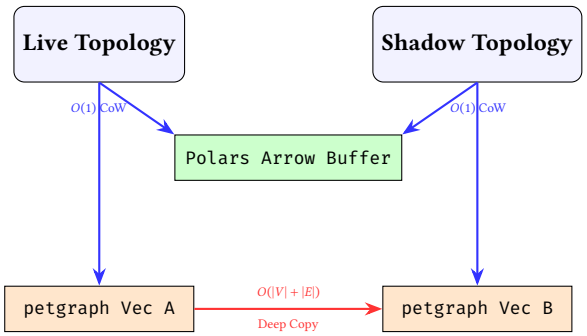


Figure 16. The Hybrid Snapshot Reality. While property DataFrames share memory instantly via CoW pointers, the underlying petgraph adjacency lists must be physically deep-copied into new memory allocations.

5.6 DeltaNet: Incremental Policy Validation

The `DeltaNetValidator` (in `nte-policy`) provides an optimised alternative to global topology validation. Rather than re-evaluating every policy against the entire graph, DeltaNet uses an **Impact Radius** algorithm to identify the minimal set of nodes affected by a mutation.

: Impact-Aware Validation

For any mutation Δ, the validator computes the affected subgraph $\mathcal{G}_\Delta \subseteq \mathcal{G}$ using a bounded BFS traversal of depth k (where k is the maximum dependency depth of the active policies). Only nodes within $\mathcal{G}_\Delta$ are re-validated, reducing validation complexity from $O(V)$ to $O(V_\Delta)$.

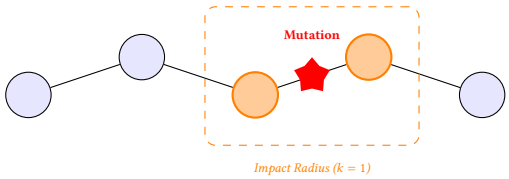


Figure 17. Incremental Validation via Impact Radius. By identifying the nodes physically and logically adjacent to a mutation, NTE avoids the $O(V)$ cost of global policy re-evaluation.

This incremental approach is essential for large-scale production networks where a single property update (e.g., a BGP weight change) should not trigger a multi-second validation sweep of 100,000 devices.

5.7 Worked Policy Examples

5.7.1 Transit Redundancy

Listing 4. Transit redundancy policy

```
# Every PoP must have at least 2 transit exit links
def check_transit_redundancy(pop_name, exit_count):
    if exit_count < 2:
        fail("PoP {} has only {} transit exits (minimum 2)".format(
            pop_name, exit_count
        ))
    return True

check_transit_redundancy(pop_name, transit_exit_count(pop_name))
```

5.7.2 Peer Diversity

Listing 5. NTE-QL policy: peer diversity

```
// P003: No two BGP sessions should share the same peer AS (from a given router)
MATCH (r:Router)-[:HAS_SESSION]->(s:BgpSession)
WITH r, s.peer_as as peer_as, count(s) as session_count
WHERE session_count > 1
RETURN r.id, peer_as, session_count
```

5.7.3 OSPF Area Consistency

Listing 6. NTE-QL policy: OSPF Area consistency

```
// P004: Ensure all interfaces in a broadcast domain share the same OSPF Area
MATCH (domain:BroadcastDomain)<-[:Parent]-(ep:Endpoint)
MATCH (ep)-[:Connectivity {layer: 'ospf'}]->(proto:OspfState)
WITH domain, count(DISTINCT proto.area_id) as area_count
WHERE area_count > 1
RETURN domain.id, area_count
```

5.7.4 MPLS Path Diversity

Listing 7. NTE-QL policy: MPLS RSVP-TE path diversity

```
// P005: Primary and Backup LSPs must be SRLG-disjoint (Shared Risk Link Group)
MATCH (p:LSP {kind: 'primary'})-[link:wire]->(target:Router)
MATCH (b:LSP {kind: 'backup', headend: p.headend, tailend: p.tailend})
WHERE p.srlg_id = b.srlg_id
RETURN p.id, b.id, p.srlg_id
```

5.7.5 No Single Point of Failure

Listing 8. No single point of failure policy

```
# All routers in the critical path must have >= 2 uplinks
def check_no_spof(critical_nodes):
    for node_id in critical_nodes:
        uplinks = check_uplinks(node_id, 2)
        if not uplinks:
            fail("Critical node {} is a single point of failure".format(node_id))
    return True

# critical_path_nodes() is a host function that returns the set of
# articulation points in the physical topology
```

```
check_no_spof(critical_path_nodes())
```

Engine Implementation (Deep Dive)

6 Storage Architecture

Storage Architecture at a Glance

Why: Combining property filtering and graph traversal efficiently requires specialised structures.
What: A dual-write system. Columnar DataFrames for fast property scans, and adjacency lists (Petgraph) for structural paths.

6.1 Rationale for Dual-Write

A graph database excels at traversal. A columnar store excels at predicate evaluation on properties. For network topology queries, you almost always need both. A query like "find all routers in the SYD PoP that have a BGP session to a peer with AS 65000" has two parts:

- *Property filter*: `pop == "SYD"`, best served by a columnar scan.
- *Structural filter*: connected via a BGP-layer edge to a peer with `as_number == 65000`, best served by graph traversal.

NTE's dual-write approach keeps both representations in sync (Figure 18) and lets the query planner pick the optimal execution order.

Figure 18 summarises how the two stores collaborate at query time.

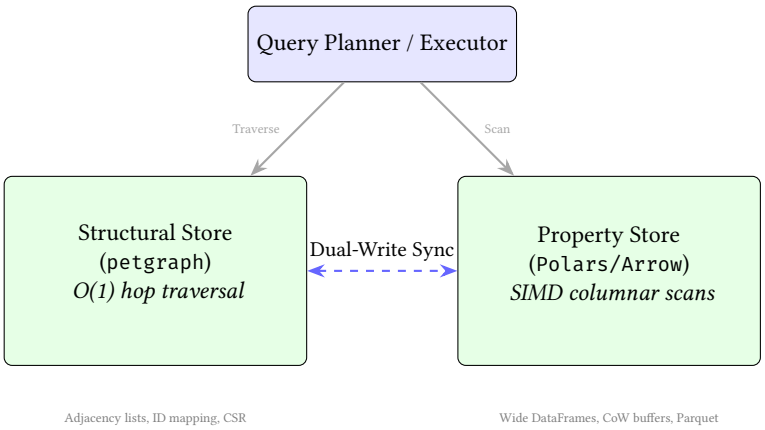


Figure 18. Dual-write architecture synergy. Structural queries target the graph engine, while property-heavy predicates target the columnar store.

6.2 petgraph StableDiGraph

The graph structure is stored in a `petgraph::StableDiGraph<GraphNode, GraphEdge>` [2]. `StableDiGraph` was chosen over the non-stable variant because:

- Node and edge indices remain valid after deletions. This is essential because NTE's ID mapping (`HashMap<i32, usize>` in both directions) must not be invalidated by unrelated mutations.
- It allows $O(1)$ lookup of node weights by `NodeIndex`, which is the primary graph-layer operation.

The `NteGraph` struct wraps `StableDiGraph` and adds:

- **Bidirectional ID mapping**: `index_map: HashMap<i32, usize>` (external → internal) and `reverse_index_map: HashMap<usize, i32>` (internal → external). All Python-facing APIs use external `i32` IDs.
- **Endpoint / node ID sets**: `HashSet<i32>` for $O(1)$ "is this ID an endpoint?" checks.
- **Layer dependency graph**: `LayerDependencyGraph` for topological ordering and layer-consistency checks.
- **Device shortcut map**: `HashMap<(i32,i32), Vec<EdgeIndex»` for fast device-to-device lookups.
- **CSR cache**: a thread-safe, lazily-computed Compressed Sparse Row representation for high-performance batch traversal (described in Section 10).

6.3 Polars Columnar Store

Node and edge properties are stored in Polars [3] DataFrames — one DataFrame per entity class (nodes, endpoints, links). Polars uses the Apache Arrow columnar format internally, which gives:

- SIMD-accelerated predicate evaluation (AVX2/AVX-512 on x86-64).
- Copy-on-Write (CoW) semantics for zero-copy snapshot cloning.
- Lazy evaluation via `LazyFrame` for filter push-down.

Properties are stored in wide columnar format (Figure 19): each unique property name becomes a column in the DataFrame. This is a deliberate trade-off. A narrow schema (one row per property) would be more flexible but would require an expensive "pivot" before columnar predicates could be applied. The wide schema means NTE scans efficiently but requires that all instances of a given node type have compatible types for a given field name.

: Sparse Columnar Compression

A hyper-scale topology generates highly heterogeneous schema (e.g., a firewall has `fw_rules`, a router has `bgp_asn`). The engine must aggressively map all unique JSON keys to a wide columnar schema. Storage explosion is avoided via Apache Arrow's validity bitmaps, ensuring that null values consume less than 1 bit per row while remaining perfectly aligned for SIMD vectorization.

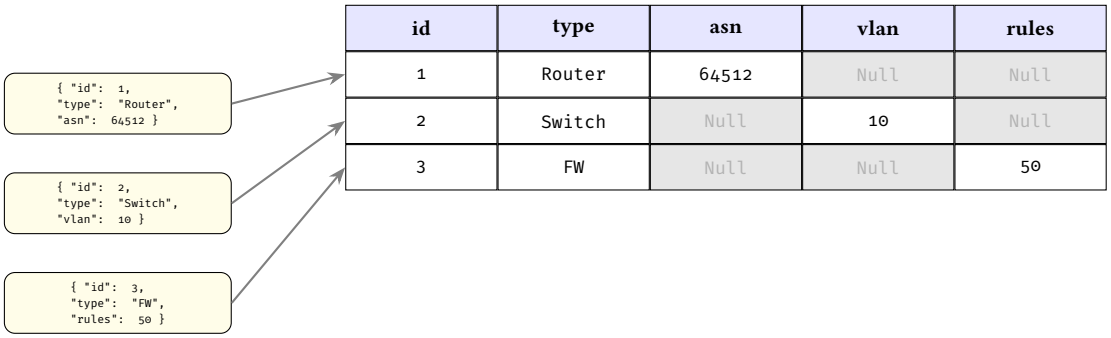


Figure 19. The Wide-Column Sparsity Matrix. Heterogeneous JSON properties are flattened into a single wide DataFrame. Arrow's validity bitmaps efficiently compress the resulting `Null` regions.

6.4 Memory Fragmentation and Defragmentation

A side effect of using `StableDiGraph` for its index stability is the creation of "tombstones" (empty slots in internal arrays) when nodes or edges are deleted. In mutation-heavy workloads, such as iterative network synthesis or large-scale failure-model simulations, these holes destroy CPU cache locality and increase memory footprint.

: Dynamic Memory Packing

To maintain hyper-scale performance, the engine must decouple *logical index stability* from *physical memory density*. NTE provides an atomic `pack()` operation that rebuilds the graph structure into a contiguous layout while preserving the isomorphism with the relational property store.

The `pack()` operation executes in $O(V + E)$ time by:

1. Allocating a new, dense `StableDiGraph` with exactly the capacity needed.
2. Mapping old node indices to new, contiguous indices.
3. Rewriting all internal state (ID maps, device shortcut caches, and edge indices embedded in weights) to reflect the new physical layout.

This operation is exposed to the Python layer as `defragment_memory()`, allowing long-running automation workers to periodically restore cache efficiency without losing data integrity.

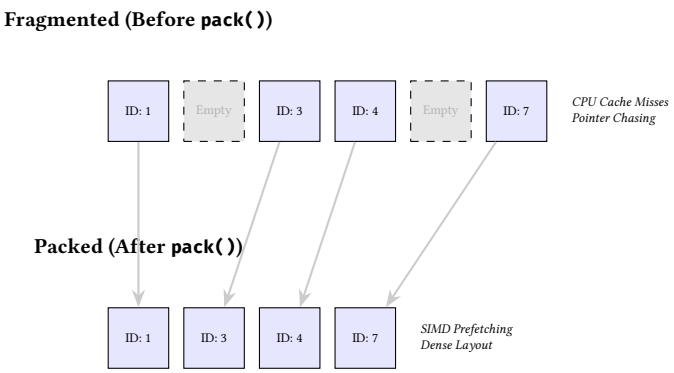


Figure 20. Memory Defragmentation (Graph Packing). The `pack()` operation removes tombstone slots from the internal adjacency arrays, restoring linear CPU cache locality and reducing memory overhead.

SIMD Vectorization. Unlike row-based graph databases that suffer from high cache-miss rates due to "pointer chasing," NTE leverages the linear memory layout of Polars to perform SIMD-accelerated predicate evaluation. By loading property columns into 512-bit registers (AVX-512), the engine can evaluate up to 16 integer comparisons in a single CPU clock cycle.

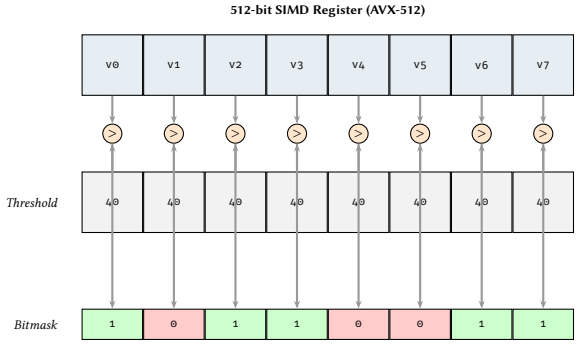


Figure 21. Hardware-Level Predicate Evaluation. A single AVX-512 instruction evaluates multiple property values in parallel, producing a bitmask that drives the subsequent graph traversal filter.

6.5 Dual-Write Consistency Protocol

Every mutation method in the `Topology` struct applies changes to both the graph and the DataFrame store before returning. The protocol is:

Algorithm 1 Dual-write mutation protocol

1: Acquire write lock on topology
2: Validate inputs (type checks, layer compatibility, ID uniqueness)
3: Apply mutation to `NteGraph` (structural side)
4: Apply corresponding mutation to `DataFrameStore` (property side)
5: If either step fails, rollback and propagate error
6: Emit `TopologyEvent` to ring buffer
7: Invalidate CSR cache
8: Release write lock

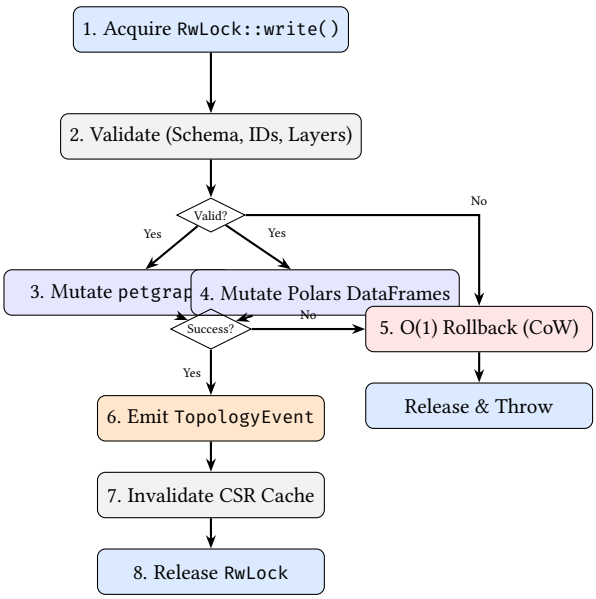


Figure 22. The dual-write transaction lifecycle. Copy-on-Write semantics allow $O(1)$ rollback if either the graph or the DataFrame mutation fails.

> **Note**
>
> NTE implements copy-on-write transaction semantics: if step 4 fails, the graph mutation in step 3 is rolled back via a compensating operation and the DataFrame store is restored from its pre-mutation Arrow CoW clone in O(1). This ensures atomicity within a single process. Durability is provided via the **Write-Ahead Log (WAL)**: every committed mutation is appended to a persistent journal before the in-memory state is updated. Following a crash, the engine can reconstruct the exact topological state by replaying the WAL events (Section 9).

6.6 Foreign Function Interface & Zero-Copy Boundaries

The engine exposes a foreign function interface (FFI) primarily via PyO3, which allows NTE to embed directly into a Python process. This approach is significantly faster than external RPC calls to a standalone graph database. To maximize concurrency and performance, NTE implements several boundary optimisations:

- **GIL Release**: The `Topology` struct is protected by a `RwLock`. Query operations acquire a read lock; mutations acquire a write lock. Crucially, the PyO3 binding layer releases the Python Global Interpreter Lock (GIL) before acquiring these Rust locks. This allows concurrent Python threads to execute native code or I/O while another thread holds a write lock in Rust.
- **Zero-Copy PyArrow**: When the `arrow-ids` feature is enabled, Polars DataFrames are shared with Python using the Arrow C Data Interface. This eliminates serialisation overhead, allowing Python data science tools to interact with NTE's columnar store seamlessly and in zero-copy fashion.

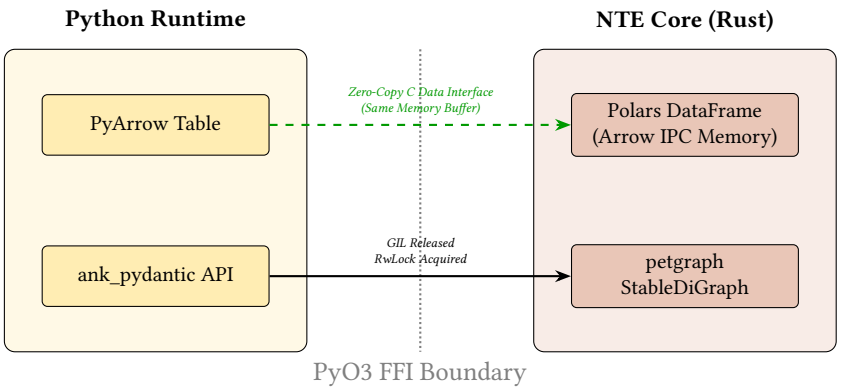


Figure 23. The FFI Zero-Copy Boundary. Unlike traditional RPC-based graph databases, NTE embeds directly in the Python process. The Arrow C Data Interface allows Python tools to read the Rust memory buffers without any serialisation overhead.

6.7 EventStore Ring Buffer

Every mutation emits a `TopologyEvent` variant into an in-memory ring buffer (a `VecDeque` managed by the `EventStore`). Events carry:

- A monotonically increasing `EventSequence` number (atomic `u64`).
- A UTC timestamp (`chrono::DateTime<Utc>`).
- Event-specific payload (node IDs, edge pairs, changed fields, etc.).

Events are serialisable to JSON (via `serde_json`) for external consumption. The ring buffer has a configurable capacity; once full, the oldest events are evicted.

6.8 Pluggable Backends

The storage layer is factored into three crates:

Crate	Description	Status
`nte-datastore-polars`	Polars Arrow-backed DataFrames	Default, production
`nte-datastore-duckdb`	DuckDB embedded OLAP database	Optional, experimental
`nte-datastore-lite`	Lightweight in-memory store	Development / testing

The `nte-datastore-core` crate defines the abstract `TopologyBackend` trait that all backends implement. Selecting a backend is a compile-time feature flag in `nte-datastore`.

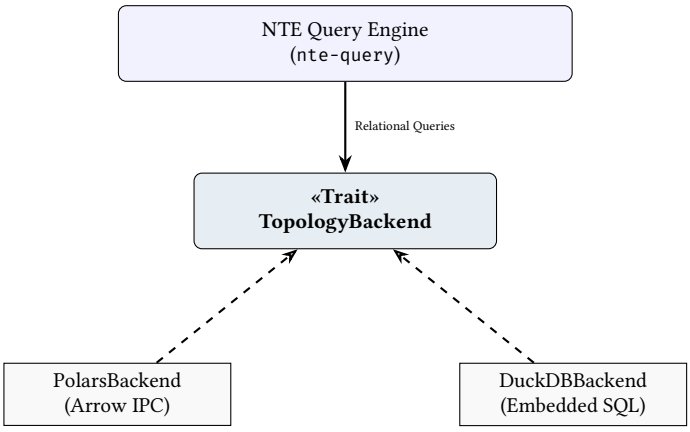


Figure 24. Trait-Based Abstraction. The core query engine interacts only with the `TopologyBackend` interface, allowing the physical columnar storage (Polars vs. DuckDB) to be swapped at compile time.

7 Algorithm Reference

This section presents the key graph and relational algorithms that underpin NTE's query planner, structural analysis, and reactive maintenance subsystems. Each algorithm is illustrated as a state-machine diagram with explicit invariants, making the correctness argument visible at a glance.

7.1 Worst-Case Optimal Joins (Leapfrog Triejoin)

Code alignment: this section mirrors `LeapfrogJoin` and `TrieIterator` in `nte-query`. The invariant "seek the minimum iterator up to max" is implemented by `LeapfrogJoin::search()` and advanced by `LeapfrogJoin::next()`.

Source

Binary joins can create massive intermediate results on cyclic patterns (triangles, cliques). NTE implements (and continues to integrate) a Worst-Case Optimal Join (WCOJ) path using Leapfrog Triejoin. Concretely, each variable binding step is reduced to a *multi-way intersection* over sorted iterators.

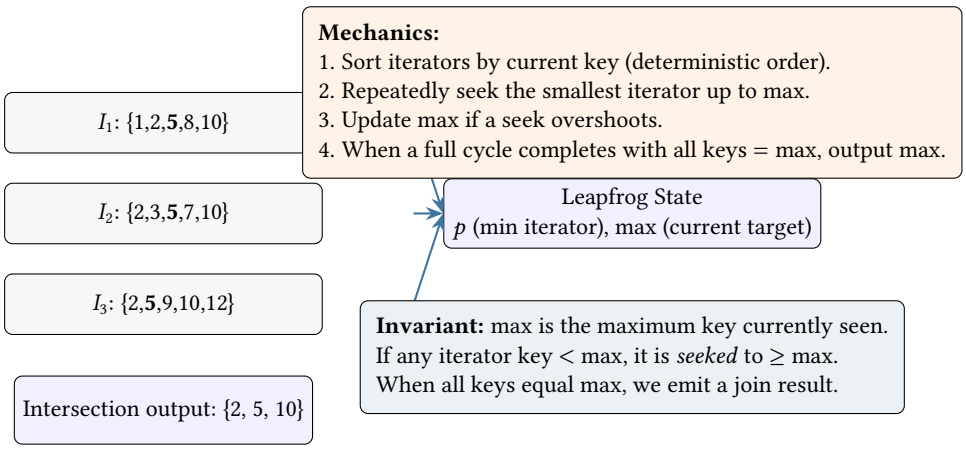


Figure 25. Leapfrog Triejoin as a stateful multi-way intersection. The algorithm maintains a single target (max) and advances the smallest iterator until all iterators align on the same key. This avoids intermediate-result explosion on cyclic patterns by intersecting all constraints simultaneously.

Why this is conceptually valuable: the join runtime is driven by iterator movement and the size of the final output, rather than by constructing and materialising large temporary relations. In NTE, the iterators are backed by CSR-sorted neighbour lists (and other filtered index views), making the intersection primitive directly usable for dense fabrics.

7.2 Betweenness Centrality (Brandes)

Code alignment: implemented as `AnalysisGraph::betweenness_centrality()` using Brandes' two-phase structure: a BFS computing `dist` and path counts `sigma`, followed by the reverse `delta` accumulation over the predecessor lists. Source

Betweenness centrality is easy to state ("fraction of shortest paths passing through a node") but non-obvious to compute efficiently. NTE uses Brandes' algorithm, which can be understood as two coupled phases per source node: a BFS that builds a shortest-path DAG and counts path multiplicity (σ), followed by a reverse accumulation of dependencies (δ).

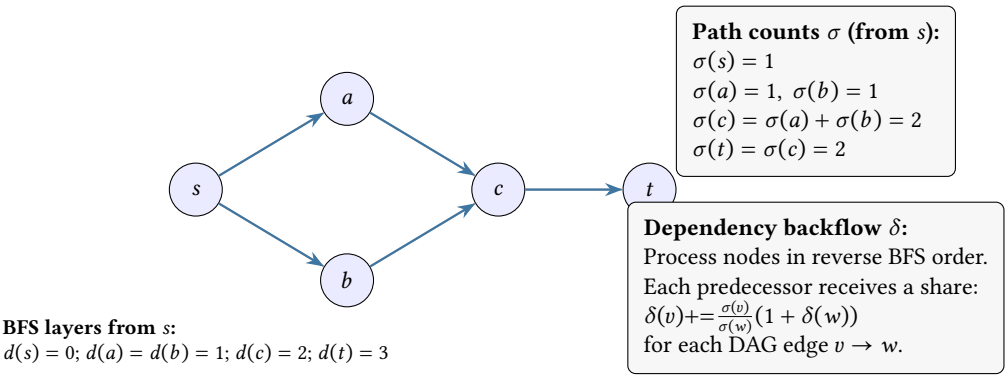


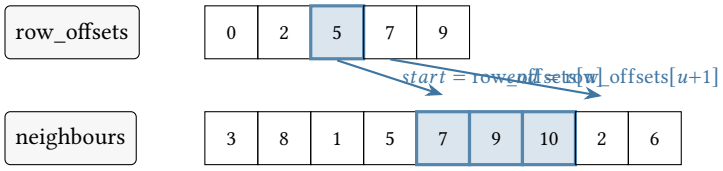
Figure 26. Brandes betweenness centrality as a conserved "flow" on the shortest-path DAG. The forward BFS counts how many shortest paths reach each node (σ). The reverse accumulation distributes dependency (δ) back to predecessors in proportion to their contribution.

Invariant to remember: for a fixed source s, the reverse pass ensures each node collects exactly the amount of "shortest-path responsibility" it carries for reaching deeper nodes. Summing this over all sources yields betweenness.

7.3 CSR Memory Layout

Code alignment: the conceptual arrays correspond to the CSR cache used for traversal in `nte-graph`. Neighbour iteration is a slice over contiguous buffers, which is also the input shape required by the WCOJ trie iterators (sorted neighbour lists). Source

The CSR (Compressed Sparse Row) representation converts pointer-heavy adjacency lists into two contiguous arrays. This matters because traversal and intersection performance is dominated by memory locality: sequential reads over dense arrays are dramatically faster than chasing per-node heap allocations.



Invariant: neighbours of a node u live in a contiguous slice neighbours$[start..end)$. This enables:
(1) BFS/DFS with sequential reads, and
(2) set intersections (e.g. WCOJ) via merge/binary-search on sorted slices.

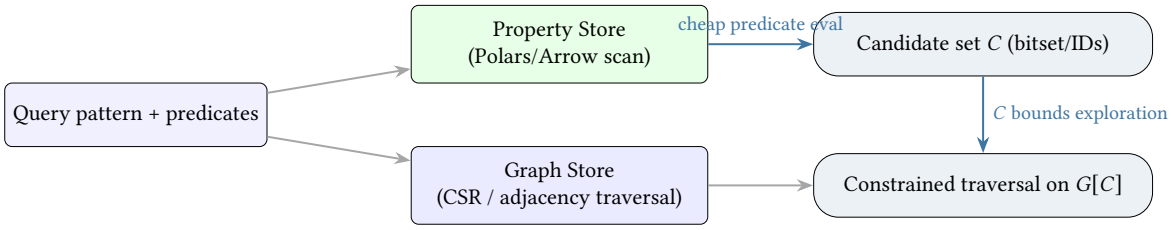
Figure 27. CSR layout. Two contiguous arrays encode the entire graph neighbourhood structure. A node's adjacency list is a slice in the neighbours array, indexed by row offsets. Locality and sorted-slice operations are the performance foundation for traversal and join-style intersections.

7.4 Filter-First Query Planning

Code alignment: the planner's costing and selectivity heuristics live in `nte-query/src/planner/` (notably `cost.rs`). The candidate-set concept corresponds to planning decisions that push property predicates earlier to reduce the induced traversal subgraph.

Source

NTE's filter-first planner turns "property predicates" into cheap candidate sets before attempting expensive structural traversal. The key idea is to make the search space explicit as a *bitset / ID set* that monotonically shrinks.



Invariant: C only shrinks (or stays equal).
Traversal cost becomes proportional to the induced subgraph.

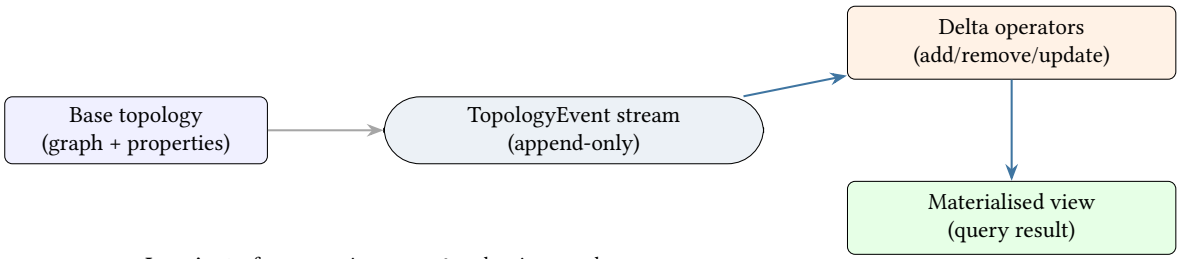
Figure 28. Filter-first planning as candidate-set narrowing. Property predicates are evaluated first to produce a set C of eligible nodes/edges. Structural traversal then runs against the induced subgraph $G[C]$, avoiding work on irrelevant regions of the topology.

7.5 Incremental View Maintenance (IVM)

Code alignment: the IVM loop consumes `TopologyEvent` values and updates its maintained state per event (see `process_event`). The "fold over events" invariant is the reason replay and continuous maintenance share the same semantics.

Source

Reactive query results can be maintained incrementally by consuming the topology's event stream. Instead of recomputing a derived view from scratch after every mutation, IVM applies small deltas to keep the view consistent.



Invariant: after processing events $0..t$, the view equals
$V_t = f(\text{Initial}) + \sum_{i=0}^{t} \Delta_i$.
This turns "continuous queries" into a deterministic fold over events.

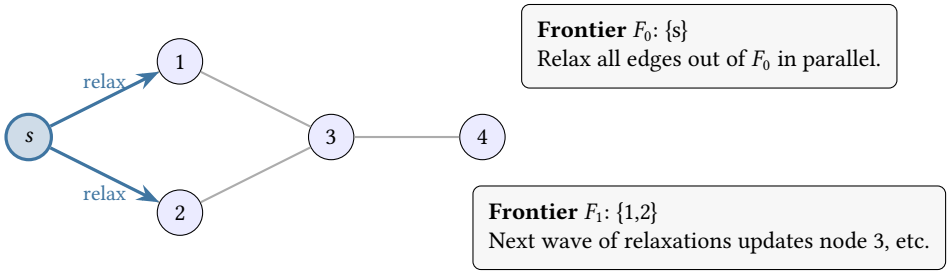
Figure 29. Incremental View Maintenance (IVM). Derived results are maintained by applying per-event deltas, rather than recomputing from scratch. This is the algorithmic bridge between the WAL/event model and reactive query semantics.

7.6 GPU SSSP as Wavefront Relaxation

Source

Code alignment: the GPU path operates over compact buffers and iterates frontiers until convergence. This section describes the execution model (bulk relaxations), not the specific kernel API.

GPU acceleration is most effective when an algorithm can be expressed as repeated bulk operations over a *frontier*. Single-Source Shortest Paths (SSSP) in this form becomes a wavefront of distance relaxations.



Invariant: distances only ever decrease; convergence when frontier empty.
The GPU advantage comes from relaxing many edges concurrently using contiguous CSR buffers.

Figure 30. GPU SSSP as wavefront relaxation. Each iteration processes a frontier of active vertices and relaxes outgoing edges in parallel. This maps naturally onto GPU execution when adjacency is stored in compact buffers.

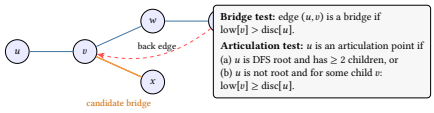
7.7 SPOF Discovery (Tarjan Bridges and Articulation Points)

Code alignment: bridge detection is implemented in `AnalysisGraph::bridges()` and articulation points in `AnalysisGraph::articulation_points()`. The diagram matches the `disc/low` propagation and the bridge predicate `low[v] > disc[u]`.

Source

To identify Single Points of Failure (SPOFs) we often need two closely related primitives: *bridges* (edges whose removal disconnects the graph) and *articulation points* (vertices whose removal disconnects the graph). NTE implements Tarjan-style DFS algorithms using two timestamps per DFS vertex:

- disc[u]: discovery time of u in DFS.
- low[u]: smallest discovery time reachable from u using *zero or more* tree edges plus *at most one* back edge.



Intuition: low[v] answers: "can this subtree escape upwards without going through its parent?" If not, the parent edge/node is a cut. In a routing context, this identifies critical physical links whose failure would partition a BGP confederation or sever an MPLS LSP.

Figure 31. Tarjan-style SPOF primitives. Discovery times (disc) order the DFS tree; low-link values (low) summarise whether a subtree has a back-edge escape route. The bridge/articulation tests are simple comparisons once low is known.

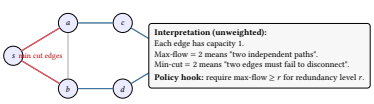
In practice, this gives a deterministic, explainable structural risk signal: "these edges/nodes are cut points". This is often more actionable than a raw connectivity boolean.

7.8 Redundancy Capacity (Max-Flow / Min-Cut)

Code alignment: the user-facing entry point is `AnalysisGraph.max_flow` in the PyO3 bindings, which delegates to the underlying analysis graph implementation. This report uses the unweighted "edge-disjoint paths" interpretation as the policy-relevant mental model.

Source

For traffic engineering and policy validation, "is there a path?" is too weak. We often want to know *how many independent ways* two nodes can stay connected. In an unweighted graph, max-flow can be interpreted as the number of edge-disjoint paths, which by Menger's theorem equals the size of a minimum cut.



Conceptual value: flow converts a qualitative statement ("redundant") into a quantitative, comparable metric that can be diffed across changes and enforced as a policy. This is used to verify OSPF multi-path availability, BGP multi-homing resilience, and MPLS RSVP-TE path diversity before committing a configuration change.

Figure 32. Max-flow/min-cut as a redundancy capacity metric. In an unweighted topology, setting each edge capacity to 1 makes the max-flow equal the number of edge-disjoint paths. The minimum cut identifies the smallest failure set that disconnects s from t.

8 Query System

Query System at a Glance

Why: Naive graph traversal is slow when properties are dense and selective.
What: A filter-first planner that drastically reduces datasets via parallel column scans before invoking structural graph traversal.

8.1 Design Goals

NTE's query system was designed around three principles:

1. **Filter-first execution**: property predicates should run before graph traversal, not after, to eliminate candidates early.
2. **Composable, typed query objects**: queries should be buildable programmatically from Python, not only via string parsing, so that tooling can construct, inspect, and serialise query plans.
3. **Honest cost estimation**: the planner should rank predicates by selectivity rather than always applying them left-to-right.

8.2 Query Representations

The `nte-query` crate exposes two parallel query representations:

`QuerySpec` A flat set of node-level filters: type, layer, kind, ID set, field equality, and expression filters. Used for "give me all nodes matching these conditions" queries. Exposed as `QuerySpec` in Python.

`QueryPlan` / `PatternNode` A structural graph pattern AST. Describes a subgraph shape to match (e.g. a router connected to a switch via an Connectivity edge). Used for multi-hop traversal queries. Exposed as `QueryPlan` and `PatternNode` in Python.

: Predicate Pushdown

In the hybrid query planner, property filters and type assertions must be pushed down to the relational store (Polars) before any structural traversal begins. This reduces the candidate node set via SIMD vectorization, effectively converting a complex $O(V + E)$ graph search into a constrained $O(K)$ bounded walk, where $K \ll V$.

8.3 Semantic Query Translation

A significant challenge in network topology databases is balancing the physical fidelity of the data model with the cognitive ergonomics of the query language. To accurately model physical failure domains (blast radius), NTE employs the strict **Endpoints Model** (see Section 3). In a naive graph database, a network is modeled simplistically as `(Router) -[CONNECTED_TO]-> (Router)`. This is insufficient for network engineering: if a router has 48 ports, the naive model cannot answer which specific port a cable is plugged into, or which connections go down if a specific line card loses power.

In NTE, physical cables and logical sessions are represented as three-hop graph traversals explicitly tracking the attachment points: `(Device) <-[Structural]- (Endpoint) -[Connectivity]-> (Endpoint) -[Structural]-> (Device)`.

However, forcing network engineers to write queries using this level of structural verbosity is counterproductive. If an engineer wants to find an eBGP session, their semantic intent is simply `(Router)-[eBGP]->(Router)`. To resolve this cognitive dissonance, NTE-QL introduces **Virtual Cypher Translation** (Figure 34). The query planner automatically bridges the gap between semantic intent and structural reality.

8.3.1 Orthogonal Layering: Layers vs. Kinds

Network engineering exists across multiple superimposed layers. A single `Connectivity` edge represents a connection *at a specific layer*. The NTE-QL grammar strictly separates the orthogonal concepts of **Layer** (the execution context, or which sub-graph to traverse) and **Type/Kind** (the specific schema class of the relationship).

Consider how different layers map to the Endpoints Model:

Example A: The Physical Layer (L1)
Nodes: `Router`, `Switch`
Endpoints: Physical transceiver ports (e.g., `GigabitEthernet0/1`)
Connection Kind: `wire` (with properties like `length`, `media_type`)
Structural Reality: `[Router:Core1] <-Structural- [Endpoint:Gi0/1] -Connectivity(wire)- [Endpoint:Te1/1] -Structural-> [Switch:Dist1]`

Example B: The Data Link Layer (L2)

Nodes: `Switch`, `Server`
Endpoints: Logical switchports or sub-interfaces (e.g., `eth0.100`)
Connection Kind: `trunk` or `access` (with properties like `vlan_id`)
Structural Reality: `[Switch:Dist1] <-Structural- [LogicalEndpoint:Vlan100] -Connectivity(trunk)- [LogicalEndpoint:eth0.100] -Structural-> [Server:Web1]`

Example C: The Routing Protocol Layer (L3+)

Nodes: `Router`
Endpoints: Loopback interfaces or routed IP addresses (e.g., `lo0`)
Connection Kind: `eBGP` or `iBGP` (with properties like `local_asn`)
Structural Reality: `[Router:Core1] <-Structural- [IP:10.0.0.1] -Connectivity(eBGP)- [IP:10.0.0.2] -Structural-> [Router:Peer1]`

8.3.2 Syntax to Data Model Mapping

To maximize expressiveness without compromising structural fidelity, NTE-QL maps its Cypher-like syntax directly to these internal constructs. In a standard graph query like `-[variable:Label {properties}]->`, the label and properties target different mechanical components in NTE:

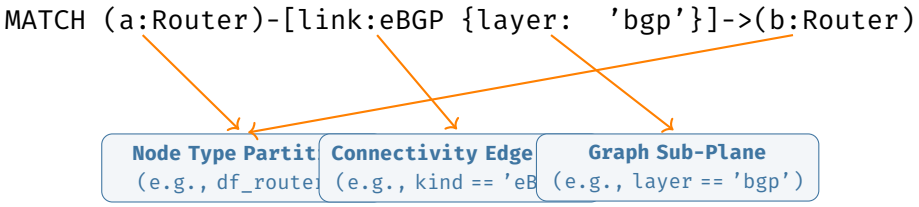


Figure 33. Syntax to Data Model Mapping. NTE-QL explicitly separates the specific relationship type (`Kind`) from the execution sub-graph (`Layer`). This allows users to query for *any* connection on the BGP layer, or specifically for *eBGP* connections across any layer.

This syntax mapping provides immense flexibility. For example:

- `MATCH (a)-[:eBGP]->(b)`: Finds `eBGP` sessions across *any* layer.
- `MATCH (a)-[{layer: 'bgp'}]->(b)`: Finds *any* connection explicitly on the `bgp` layer.
- `MATCH (a)-[:eBGP {layer: 'bgp'}]->(b)`: Finds specifically `eBGP` sessions on the `bgp` layer.

8.3.3 Virtual Cypher Expansion

When a user submits a query, they do not need to specify the `Structural` edges or the `Endpoint` nodes unless they are explicitly querying port-level properties.

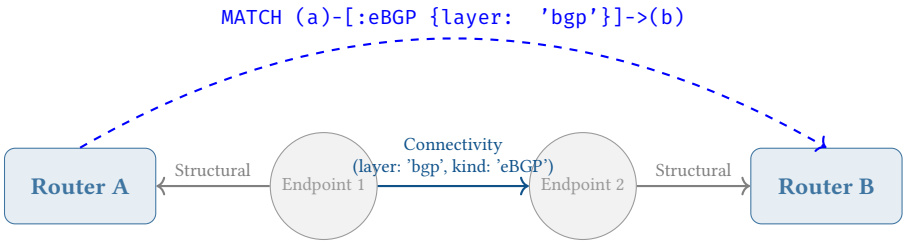


Figure 34. Virtual Cypher Translation. The user specifies a semantic device-to-device relationship (dashed line). The query planner intercepts this and automatically compiles it into the three-hop endpoint traversal (solid lines) required by the structural data model.

The NTE-QL mapping rules are as follows:

- **Cypher Node Label** (e.g., `:Router`): Maps to the `node_type` in the partitioned Polars DataFrame.
- **Cypher Edge Label** (e.g., `:eBGP`): Maps to the `kind` property of the underlying `Connectivity` edge.
- **Cypher Edge Property** (e.g., `{layer: 'bgp'}`): Defines the sub-graph boundary, instructing the planner to strictly traverse `Connectivity` edges belonging to the BGP partition.

For example, to find all active eBGP sessions, the user writes:

```
MATCH (a:Router)-[session:eBGP {layer: 'bgp', status: 'active'}]->(b:Router)
RETURN a.hostname, b.hostname
```

The Query Planner intercepts the Abstract Syntax Tree (AST). It detects that an edge was drawn directly between two entity types known to be Devices (or detects the absence of explicit `Endpoint` nodes). The `PatternExecutor` automatically compiles the `(a)-[:eBGP]->(b)` AST node into a multi-step execution plan: Find `Router` nodes → Follow incoming `Structural` edges to their endpoints → Follow outgoing `Connectivity` edges on the `bgp` layer where `kind == 'eBGP'` and `status == 'active'` → Follow `Structural` edge to target `Router`.

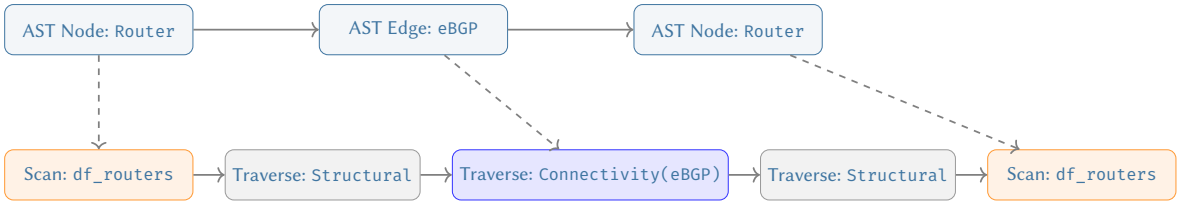


Figure 35. AST Compilation to Execution Plan. The semantic Cypher AST (top) is mathematically expanded by the Query Planner into a deterministic traversal plan (bottom) across the normalised Endpoints graph and partitioned DataFrames.

8.3.4 Parallel Expansion: The LAG Scenario

The Virtual Cypher abstraction becomes particularly valuable when dealing with complex, multi-homed connectivity like Link Aggregation Groups (LAGs) or Equal-Cost Multi-Path (ECMP) routing.

In a physical data center, two core routers might be connected via four separate 100G fibre cables (a 400G LAG bundle). In the NTE Endpoints Model, this is explicitly represented as four distinct `Connectivity` edges connecting four distinct pairs of `Endpoint` endpoints.

However, when an engineer queries `MATCH (a:Router)-[link:LAG]->(b:Router)`, they typically want to treat that bundle as a single logical connection. The NTE-QL planner handles this via **Parallel Expansion**. Because it understands the structural semantics, the planner can collapse the four parallel `Connectivity` paths back into a single semantic relationship, aggregating the underlying endpoint properties (e.g., summing the bandwidth) before returning the result to the user.

8.3.5 Context Projections and Sub-Selects

To avoid repeating `{layer: '...'}` in complex queries, NTE-QL employs a context scope mapping via the `USE LAYER` directive. This acts as a global namespace projection, instructing the planner to strictly traverse edges belonging to that topological plane for the duration of the query.

```
USE LAYER 'physical'
MATCH (a:Switch)-[link:trunk]->(b:Switch)
```

This layer abstraction is particularly powerful for **Cross-Layer Queries**, where a user wants to find how logical states depend on physical infrastructure.

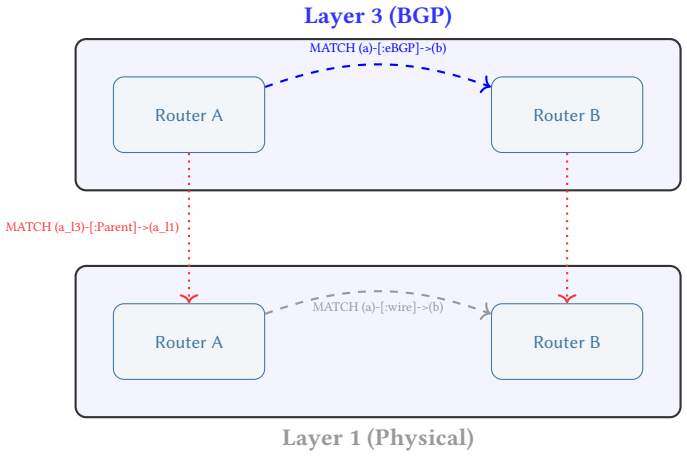


Figure 36. Cross-Layer Semantic Querying. Users can independently query the semantic graph of each layer and trace dependencies across them using the `Parent` edges, entirely abstracted from the underlying port complexities.

```
// Cross-Layer Query Example
MATCH (r1:Router)-[session:eBGP {layer: 'bgp'}]->(r2:Router)
MATCH (r1)-[:Parent]->(r1_phys)
MATCH (r2)-[:Parent]->(r2_phys)
```

```
MATCH (r1_phys)-[cable:wire {layer: 'physical'}]->(r2_phys)
RETURN session.status, cable.media_type
```

8.3.6 Bridging the Gap: The Benefits of Virtual Cypher

The combination of the Endpoints Model and Virtual Cypher Translation yields a best-of-both-worlds architecture:

1. **Mathematical Purity for Algorithms**: When graph algorithms like Shortest Path (CSPF) or PageRank execute, they operate on the pure, fully materialized Endpoints Model. They understand exactly which line cards represent single-points-of-failure and which cables share physical conduits.
2. **Ergonomic Queries for Humans**: Network engineers do not need to memorise the internal database schemas or write repetitive `Structural` loops. They can express their declarative intent natively: "Find all switches running OSPF connected to this core router."
3. **Zero-Copy Overhead**: Because the translation happens entirely within the AST compiler phase of the Query Planner, there is zero runtime overhead. No "shortcut" edges are generated, stored, or synchronised in the database.
4. **Dense Storage Partitioning**: Because the semantic labels translate directly into DataFrame types, the execution engine can partition the data intelligently. Queries for `:Router` immediately skip the memory blocks containing millions of `:Endpoint` or `:Server` records.

This planner-driven translation guarantees that the underlying graph remains highly normalised for strict mathematical blast-radius simulation, while the human-facing query language remains declarative, beautiful, and domain-aligned.

8.3.7 Programmatic Mapping: Python and Starlark

While NTE-QL provides a declarative string interface, NTE is designed to be embedded within larger automation systems. Therefore, these exact semantic translation concepts map directly into the programmatic host languages: the **Python SDK** (for external orchestration) and the embedded **Starlark Policy Engine** (for internal validation).

Python SDK Execution In the Python SDK, queries can be built programmatically using the fluent builder API. The Virtual Cypher translation is identical: the Python user asks for a device-to-device relationship, and the Rust core executes the three-hop endpoint traversal.

```
# Find all core routers connected via 100G trunks
query = (
    topo.query()
    .match_node("core", node_type="Router", role="core")
    .match_edge("link", edge_type="trunk", layer="physical", bandwidth="100G")
    .match_node("peer", node_type="Router", role="core")
)
results = query.execute()
```

Starlark Policy Invariants NTE embeds a Starlark (Python dialect) runtime to evaluate strict design contracts against the topology before allowing mutations to commit. Because Starlark is a sandboxed language, it cannot execute arbitrary string queries. Instead, it relies on these semantic graph projections exposed via the context bindings.

When a Starlark policy checks an invariant (e.g., "all edge switches must have redundant uplinks"), it uses the same semantic mapping abstractions to effortlessly traverse the underlying normalised physical graph.

```
# Starlark Policy: Verify Redundant Uplinks
def validate_uplinks(ctx):
    # Context bindings automatically abstract the Structural/Connectivity hops
    edge_switches = ctx.find_nodes(type="Switch", layer="physical", role="edge")

    for switch in edge_switches:
        # get_neighbors automatically traverses the 3-hop physical connection
        uplinks = switch.get_neighbors(edge_type="wire", layer="physical")
        if len(uplinks) < 2:
            ctx.fail("Switch {} lacks redundant uplinks", switch.id)

    return ctx.findings()
```

By unifying the NTE-QL Cypher semantics, the Python SDK, and the Starlark Policy Engine around this Virtual Translation layer, NTE provides a consistent, high-level abstraction across all programmatic interfaces without sacrificing the structural integrity of the Endpoints Model.

8.4 Execution and Optimisation

The query compilation pipeline (Figure 40) translates high-level NTE-QL strings or Python `QueryPlan` objects into executable filter and traversal plans.

8.4.1 Query Compilation and Execution Planning

Before execution, high-level NTE-QLstrings or Python `QueryPlan` objects must be translated into an executable form. This process is known as **Query Compilation**. It is during this compilation phase that the **Virtual Cypher Translation** (described in Section 8.3) is instantiated.

The compilation process decomposes the semantic query into its relational and structural components:

1. **AST Generation**: The NTE-QL string (e.g., `MATCH (a:Router)-[:eBGP]->(b:Router)`) is parsed into a high-level Abstract Syntax Tree (AST).
2. **Semantic Expansion**: The Query Planner recognises the device-to-device relationship. It mutates the AST, inserting the necessary `Structural` and `Connectivity` hops to satisfy the physical Endpoints Model constraints.
3. **Filter Compilation**: Cypher labels (like `:Router`) and inline properties are stripped from the AST and compiled into Polars `LazyFrame` execution plans.
4. **Structural Compilation**: The remaining expanded AST (representing the 3-hop traversal) is compiled into a `petgraph` finite state machine.

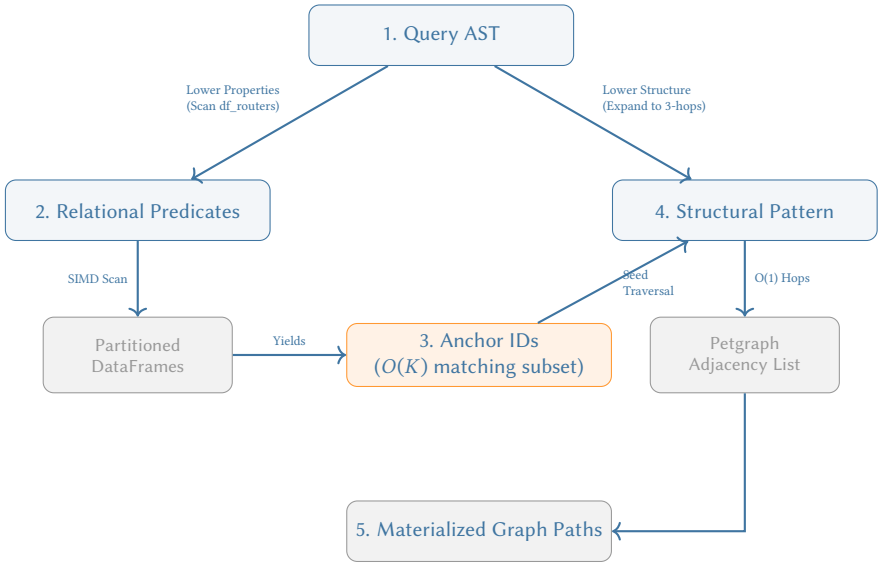


Figure 37. The Filter-First Hybrid Execution Pipeline. During the AST compilation phase, the engine splits the query. It executes highly-selective property predicates against partitioned columnar storage *first*, yielding a tiny subset of Anchor IDs. These anchors seed the expanded 3-hop structural graph traversal, guaranteeing that the engine never traces paths from irrelevant nodes.

The planner uses a **Cost-Based Execution Planning** strategy: if a query contains highly selective property filters (e.g., a specific `hostname`), the relational scan is executed first to prune the graph search space. If the query is purely structural, it is compiled directly for the petgraph engine.

8.4.2 Query Complexity: SIMD Pruning vs. Adjacency Traversal

The performance advantage of NTE's hybrid approach can be quantified using Big-O analysis. Consider a property-heavy structural query (e.g., "Find all Routers where pop='SYD' and follow their 1-hop links").

Naive Traversal cost $O(V+E)$. In a row-based graph store, we must visit every node V to check the property, then traverse its adjacency list E. This is dominated by pointer chasing and cache misses.

NTE Filter-First cost $O(\frac{V}{K} + E_{sub})$. where K is the SIMD vectorization factor ($K = 16$ for AVX-512) and E_{sub} is the set of edges connected only to the matching nodes.

Since $E_{sub} \ll E$ in selective queries, the query time is reduced by approximately an order of magnitude. The relational scan $O(V/K)$ is a linear scan of contiguous memory, maximizing cache locality and CPU throughput.

8.4.3 QuerySpec: The Relational Path

The `QueryExecutor` in `nte-query` compiles a `QuerySpec` into an ordered `FilterPlan` — a sequence of `FilterOp` variants sorted by `FilterRank`:

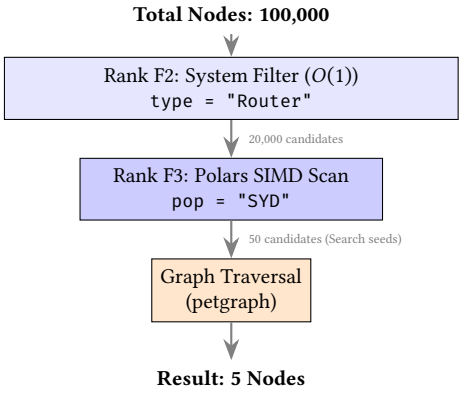


Figure 38. The Cardinality Funnel (Filter-First Execution). Cheap columnar operations aggressively prune candidates, ensuring the more expensive graph traversal engine only visits a tiny fraction of the topology.

Execution proceeds left-to-right. Each filter receives the candidate set from the previous filter and returns a (potentially smaller) subset. The Polars LazyFrame pipeline is constructed in-order and materialised with a single `collect()` call, benefiting from Polars' internal optimiser.

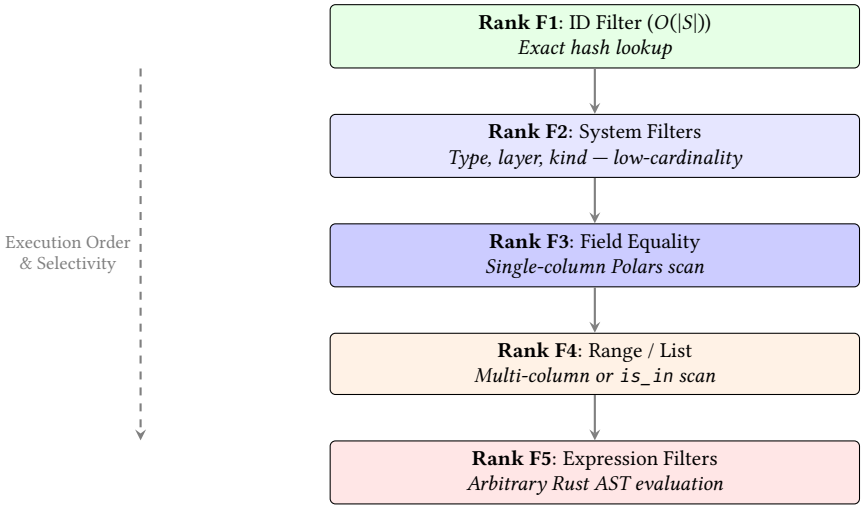


Figure 39. Query Planner Execution Ranks. Predicates are sorted and executed from cheapest (F1) to most expensive (F5) to continuously reduce the candidate set.

Filter plan example

Query: "All routers in the SYD PoP with AS number 64512"

1. F2: type = "Router" → scan `node_types` DataFrame
2. F3: `pop == "SYD"` → filter `data_pop` column
3. F3: `as_number == 64512` → filter `data_as_number` column

Polars executes steps 2 and 3 as a single lazy pipeline after step 1 narrows the candidate set.

8.4.4 Pattern Matching

The `PatternNode` AST describes the shape of a subgraph. Supported patterns:

`PatternNode::node(type)` Match a single vertex of the given type.

`PatternNode::binding(name, type)` Match a vertex and bind it to a name for later reference.

`PatternNode::chain(steps)` A sequence of alternating node-steps and edge-steps.

`PatternNode::union(patterns)` Union of multiple patterns.

Variable-length paths `HopSpec::exact(n)`, `HopSpec::range(min,max)`, or `HopSpec::any()`.

The pattern matcher performs a depth-first search over the graph, pruning branches when a node fails the pattern constraint. The maximum number of matches is capped at `QueryPlan::DEFAULT_MAX_MATCHES` (10,000) by default to prevent runaway traversals; the result carries a `truncated` flag when this cap is reached.

8.4.5 Selectivity Ranking in Detail

The five filter ranks correspond to the following `FilterOp` variants:

ID Filtering (`FilterRank::Id`) — `HashSet<i32>` lookups are the fastest and always execute first.

System Filtering (`FilterRank::System`) — Queries on Type, Layer, or Kind use low-cardinality columns.

Equality Filtering (`FilterRank::Equality`) — Single-field property equality scans.

Range Filtering (`FilterRank::Range`) — Set membership (`IN`) or numerical range checks.

Expression Filtering (`FilterRank::Expression`) — Arbitrary `ExprNode` AST evaluations, always executed last.

Field equality filters on list values are promoted from `Equality` to `Range` rank because they require a Polars `is_in` operation, which is costlier than a direct equality predicate.

8.4.6 Query Profiling

Setting `QueryHints{enable_profiling: true}` in a `QuerySpec` records timing data for each filter step. The profiler captures per-step candidate counts and wall-clock durations, enabling identification of expensive filters. The profile is returned alongside the result set.

8.4.7 Plan Cache

NTE maintains an LRU plan cache (capacity configurable, default 256 entries) for `QuerySpec` objects. The cache key is a `CacheKeyMaterial` struct containing sorted, canonicalised representations of all filter fields (to avoid cache misses from HashMap ordering non-determinism). Cache keys are computed lazily via `OnceLock` to avoid repeated sorting allocations.

> **Note**
>
> The plan cache caches compiled `FilterPlan` objects, not query results. Result caching (returning previously computed row sets) is available via `QueryHints{cache_results: true}` but is off by default, as topology mutations must invalidate the result cache.

8.5 User-Facing Interfaces

NTE exposes the query engine through multiple interfaces, each targeting a different user persona and integration context.

8.5.1 Query API Decision Matrix

NTE provides three distinct query paradigms. To avoid fragmentation, callers should select their interface based on the following matrix:

Interface	Primary Audience	When to use
Programmatic (`QuerySpec`)	Integrators	Internal system-to-system queries where parameters are highly dynamic or built from ASTs.
Fluent Builder	Data Scientists	Interactive Python notebooks; chaining complex property filters.
NTE-QL (Cypher)	Network Engineers	Human-readable graph pattern matching, CLI/TUI usage, and external dashboarding.

8.5.2 NTE-QL String Interface

In addition to the programmatic query API, NTE supports a Cypher-inspired string query language called NTE-QL. Queries are parsed by a PEG grammar implemented with the `chumsky` parser combinator library and compiled into `QueryPlan` / `QuerySpec` objects before execution.

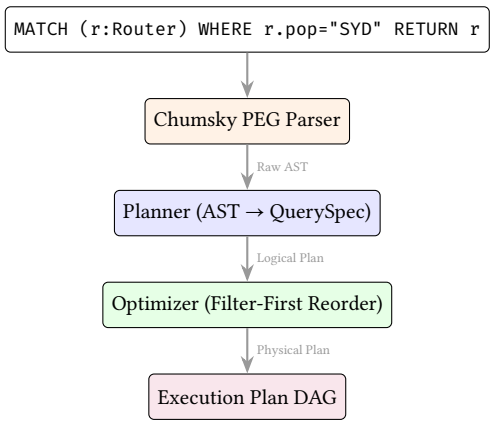


Figure 40. The NTE-QL compilation pipeline. String queries are parsed, converted into programmatic `QuerySpec` structures, and fed into the same optimizer as the Python APIs.

The grammar supports a variety of expressions. Here is a brief survey of supported query patterns:

Simple node scan

```
MATCH (r:Router)
WHERE r.pop = "SYD" AND r.as_number = 64512
RETURN r.hostname
```

Two-hop traversal: direct BGP peers

```
MATCH (r:Router)-[:Connectivity]->(s:Router)
WHERE r.hostname = "core-01"
RETURN r.hostname, s.hostname, s.as_number
```

Bounded-hop reachability: within 3 hops

```
MATCH (src:Router)-[*1..3]->(dst:Router)
WHERE src.hostname = "edge-01"
RETURN dst.hostname
```

Aggregation: node degree

```
MATCH (r:Router)-[:Connectivity]->(s:Router)
WHERE r.pop = "SYD"
RETURN r.hostname, COUNT(s) AS peer_count
ORDER BY peer_count DESC
```

NOT / exclusion: isolated routers

```
MATCH (r:Router)
WHERE NOT EXISTS {
    MATCH (r)-[:Connectivity]->(:Router)
}
RETURN r.hostname
```

Path finding: shortest path

```
MATCH p = shortestPath((src:Router)-[*]->(dst:Router))
WHERE src.hostname = "r1" AND dst.hostname = "r4"
RETURN [n IN nodes(p) | n.hostname] AS path
```

8.5.3 The ExprNode AST

Complex filter predicates are represented as an `ExprNode` abstract syntax tree. The full set of node types:

Variant	Description
`Field(name)`	Reference to a node/edge property
`BindingField(binding, name)`	Property access via a named binding
`Literal(value)`	Constant (null, bool, int, float, string, list)
`Not(expr)`	Logical negation
`Logical(op, left, right)`	AND / OR
`Comparison(op, left, right)`	$=, \neq, <, \leq, >, \geq$
`Arithmetic(op, left, right)`	$+, -, \times, \div$
`StringOp(op, field, pattern)`	contains, startsWith, endsWith, matches
`IsNull(expr)` / `IsNotNull`	Null checks
`IsIn(field, values)`	Membership test

8.5.4 Python Fluent Query Builder

In addition to the programmatic `QuerySpec` / `PatternNode` APIs and the NTE-QLstring interface, NTE provides a fluent Python query builder that does not require `ank_pydantic`. The builder is inspired by Polars' expression API and compiles down to the same `QuerySpec` objects used by the Rust executor.

Fluent query builder: node queries

```python
from ank_nte import Topology
from ank_nte.query import QueryNamespace, Expr

t = Topology()
# ... populate topology ...
q = QueryNamespace(t)

# Chainable node query
syd_routers = (
    q.nodes()
     .of_type("Router")
     .in_layer("physical")
     .filter(Expr.field("pop") == "SYD")
     .sort("hostname")
     .ids()
)

# Expression DSL supports operators
fast_ports = (
    q.nodes()
     .of_type("Endpoint")
     .filter(Expr.field("speed_gbps") > 40)
     .count()
)

# String operations
juniper_devices = (
    q.nodes()
     .of_type("Router")
     .filter(Expr.field("vendor").contains("Juniper"))
     .collect()
)
```

Fluent query builder: link queries

```python
# Link queries for edge filtering
transit_links = (
    q.links()
     .of_type("Connectivity")
     .in_layer("physical")
```

```
    .filter(Expr.field("speed_gbps") >= 100)
    .ids()
)

# Between-nodes filtering
r1_links = (
    q.links()
    .between(source_ids=[1], target_ids=[2])
    .collect()
)
```

The expression DSL supports the full `ExprNode` AST via Python operator overloading: `==`, `!=`, `>`, `>=`, `<`, `<=` for comparisons; `&` (and), `|` (or), `~` (not) for logic; `+`, `-`, `*`, `/` for arithmetic; and methods `.contains()`, `.startswith()`, `.endswith()`, `.matches()` for string operations. Terminal methods include `.ids()`, `.collect()`, `.count()`, `.exists()`, `.first()`, and `.one()`.

8.5.5 Query Plan Visualisation

NTE can render query execution plans as Mermaid flowchart diagrams. This is useful for understanding how the planner decomposes a query and for identifying optimisation opportunities. The `nte-query` crate provides a recursive generator that traverses the `ExecutionPlan` tree to emit Mermaid flowchart syntax, mapping `NodeScan`, `EdgeScan`, `WcojJoin`, and `SemiJoin` operations to visual blocks.

The visualisation is accessible in two ways:

1. **NTE-QL REPL**: `EXPLAIN VISUAL <query>` renders the plan inline.
2. **HTTP API**: `GET /explain?query=...` returns an HTML page with the rendered Mermaid diagram (via the `nte-server` diagnostics endpoint).

8.5.6 NTE-QL Interactive REPL

The `nte-cli` crate provides a standalone command-line interface with three modes:

`nte-cli repl` An interactive NTE-QL shell (built on `reedline`) that parses queries, compiles them to execution plans, and displays results in formatted tables (via `comfy_table`). The `EXPLAIN` prefix shows the compiled plan; `EXPLAIN VISUAL` renders the Mermaid DAG.

`nte-cli validate` A policy validation workflow: loads a YAML policy file and a topology archive, evaluates all policies, and renders a compliance report.

`nte-cli dashboard` A TUI dashboard (built on `ratatui` and `crossterm`) showing real-time memory usage, CPU load, graph statistics (node/edge counts), and a 30-entry history sparkline.

NTE-QL REPL session

```
nte> MATCH (r:Router) WHERE r.pop = "SYD" RETURN r.hostname
+----------+
| hostname |
+----------+
| r-syd-01 |
| r-syd-02 |
+----------+
2 rows returned

nte> EXPLAIN VISUAL MATCH (r:Router)-[:Connectivity]->(s:Router) RETURN r, s
-- Mermaid DAG rendered --
```

8.6 End-to-End Query Trace

To illustrate the filter-first query planner's coordination between the relational and graph stores, we trace a representative multi-stage query (Figure 41):

> *"Find all spine routers in the SYD PoP with AS 64512 that have a < 3 hop path to a firewall."*

The query execution follows three logical phases, as shown in Figure 41.

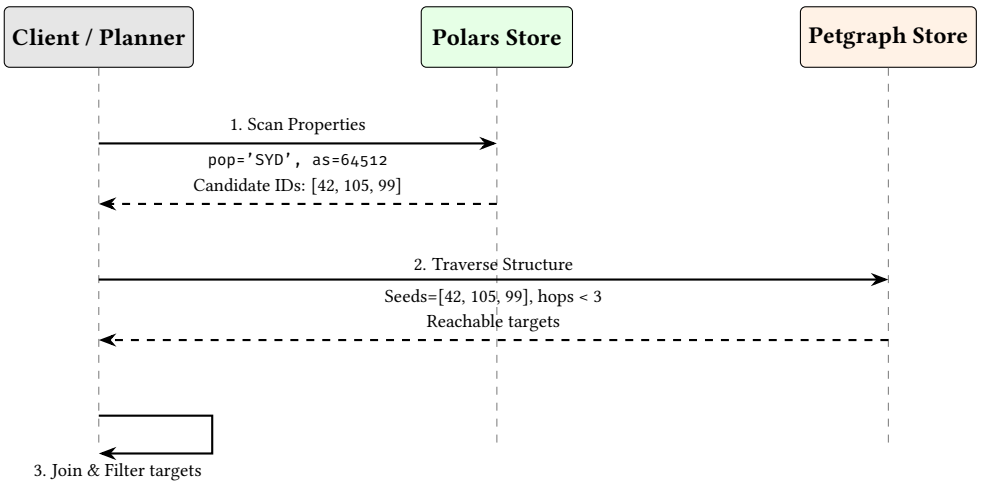


Figure 41. The Filter-First Pipeline. Relational scans prune the search space before expensive graph traversals are initiated.

Insight:
By using the Polars store to prune 99% of nodes based on properties (`pop`, `as_number`), the subsequent graph traversal only needs to visit a tiny fraction of the total adjacency list.

Operational Trace: What Actually Runs

In a real execution, Phase 1 emits a candidate ID vector from the columnar store. That vector becomes the seed set for Phase 2, where the traversal is bounded by a hop limit and early-terminated by predicate checks. Phase 3 performs a final join back to relational properties to render result rows and to produce an explainable plan trace for `EXPLAIN`.

9 Transactions

9.1 The Dual-Write Transaction Model

NTE's defining architectural feature is its **Hybrid Storage Architecture** (Section 6), which pairs a `petgraph` structural adjacency list with `Polars` columnar DataFrames. While this provides extraordinary query performance (as proven in Section 8.4.1), it introduces a severe data consistency challenge: how do you guarantee that the graph structure and the tabular properties never fall out of sync?

NTE solves this via the **Dual-Write Transaction Model** (Figure 42). Every mutation command submitted to the engine is wrapped in an atomic transaction scope that strictly enforces consistency across both underlying storage engines.

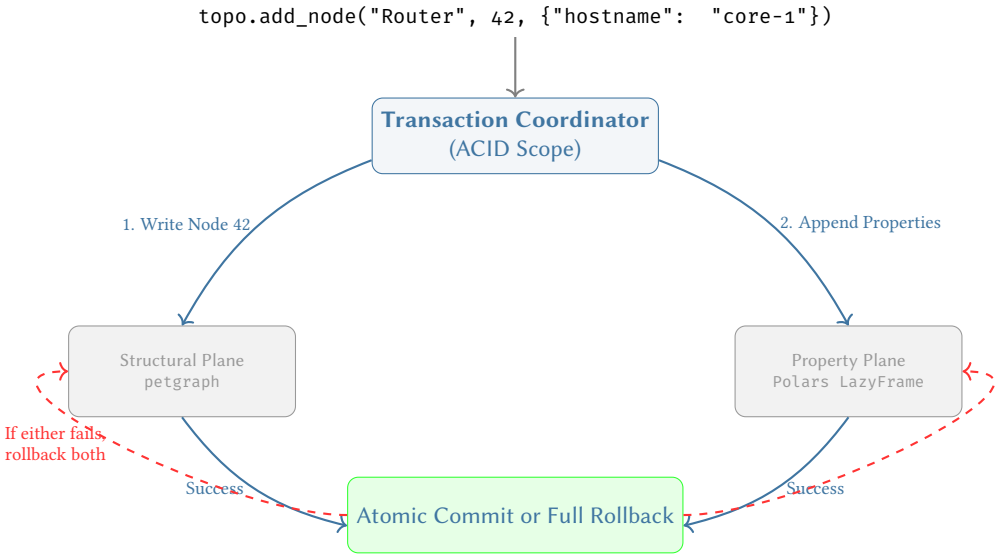


Figure 42. The Dual-Write Transaction Model. Mutations are coordinated atomically across the divergent storage engines. If a node is added to the graph but the properties fail schema validation in Polars, the entire transaction aborts, ensuring the digital twin never enters a split-brain state.

The transaction model utilises Arrow's reference-counted Copy-on-Write (CoW) memory buffers for $O(1)$ snapshot creation at transaction start, and $O(1)$ rollback if the transaction is aborted.

9.2 ACI (Durability via WAL) Transaction Model

Transaction Durability

NTE implements full **ACID** transactions (Atomicity, Consistency, Isolation, Durability). While structural mutations utilise copy-on-write (CoW) semantics for isolation, persistent durability is guaranteed by an optional **Write-Ahead Log (WAL)**. Every committed mutation is appended to a monotonically increasing journal on disk, ensuring that in-memory state can be recovered following a process crash.

Every mutation can be wrapped in a transaction scope that atomically commits or rolls back changes to both the graph and the DataFrame store.

The transaction model (Figure 43) uses Arrow's reference-counted CoW buffers for $O(1)$ snapshot creation at transaction start and $O(1)$ rollback if the transaction is aborted. When WAL is enabled, the commit phase is a two-step process:

1. **Prepare**: The mutation is validated and prepared in-memory.
2. **Commit**: The event is flushed to the WAL on disk, then applied to the primary in-memory stores.

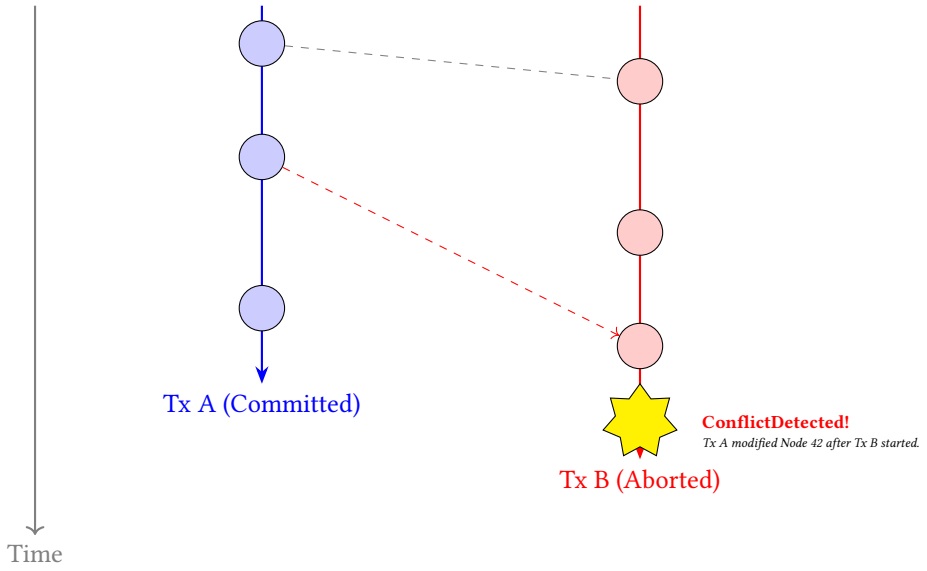


Figure 43. MVCC Transaction Timeline. Both transactions read from the same initial snapshot S_0. Because Tx A commits a mutation to Node 42 before Tx B attempts to mutate it, the Conflict Engine safely aborts Tx B to prevent a Write-Write conflict.

9.3 Isolation Levels

Three isolation levels are supported, matching standard database semantics:

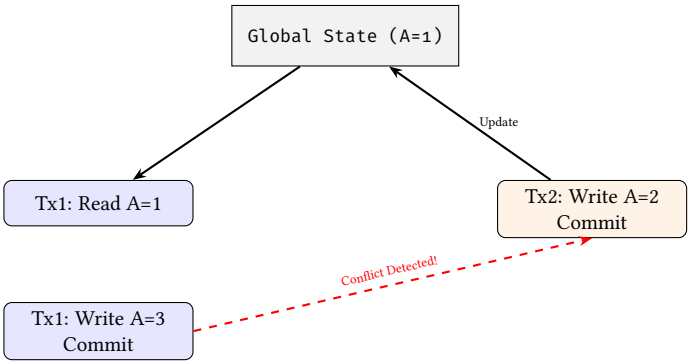


Figure 44. Serializable isolation conflict detection. Tx1 fails to commit because its read-set (A=1) was invalidated by Tx2's concurrent write.

Level	Behaviour
`ReadCommitted`	Sees only committed data; each read within the transaction sees the latest committed state.
`RepeatableRead`	Reads within the transaction see a consistent snapshot taken at transaction start.
`Serializable`	Full conflict detection: transactions that read/write overlapping data are detected and the later transaction is aborted.

9.4 Transaction Snapshots

NTE's copy-on-write architecture allows the engine to expose consistent, read-only snapshots of the topology at any point in time. Each transaction creates a logical snapshot at its start time, using persistent data structures for structural sharing.

The use of immutable persistent hash maps and reference-counted DataFrames means that a snapshot shares most of its structure with the live topology. Only modified entries are copied, giving near-constant memory overhead for transactions that touch a small fraction of the graph.

9.5 Python Transaction API

The `PyTransaction` class exposes transactions as a Python context manager:

Listing 9. Transaction API

```python
from ank_nte import Topology, IsolationLevel

t = Topology()
# ... populate topology ...

# Context manager: auto-rollback on exception
with t.begin_transaction(IsolationLevel.Serializable) as txn:
    t.add_nodes(ids=[100], node_types=["Router"], layer="physical",
                data=[{"hostname": "r-new"}])
    t.add_inter_edges(sources=[10], destinations=[100])
    txn.commit()  # explicit commit

# If an exception occurs before commit(), the transaction
# is automatically rolled back (both graph and DataFrame store).

# Manual rollback
with t.begin_transaction() as txn:
    t.remove_node_cascade(node_id=1)
    if not t.is_reachable(source=2, target=3):
        txn.rollback()  # revert: removal would partition the graph
    else:
        txn.commit()
```

Durable Commit

NTE transactions can be made crash-consistent by enabling the optional Write-Ahead Log (WAL). When active, every committed transaction is flushed to a persistent on-disk journal before the in-memory state is updated. This ensures that the engine can fully recover its state even following a hardware failure or process crash.

9.6 Conflict Detection

When multiple transactions execute concurrently, NTE detects conflicts using a `VersionTracker` that records per-node and per-edge version numbers. At commit time, the tracker compares each transaction's read and write sets against committed versions:

Write–write conflicts (all isolation levels): if a node or edge was modified by another transaction after the current transaction began, a `ConflictError` is raised.

Read–write conflicts (Serializable only): if a node that was *read* by the current transaction was subsequently *written* by another committed transaction, a `ConflictError` is raised. Read-your-own-writes are excluded from this check.

Conflict errors carry the transaction ID, the list of conflicting node/edge IDs, and the conflict type (`WriteWrite` or `ReadWrite`), allowing callers to implement application-level retry or merge logic.

10 Persistence and Snapshots

: Zero-Copy Isolation

Snapshots must leverage Arrow's reference-counted, memory-mapped buffers to guarantee that historical state requires zero initial memory overhead. Cloning the property schema is an $O(1)$ pointer increment. This isolation underpins the engine's transactional rollbacks and "What-If" shadow topologies.

10.1 Named Snapshots

NTE supports named in-memory snapshots managed by `SnapshotManager`. The key implementation detail is that Polars DataFrames use Apache Arrow's reference-counted buffers. Cloning a DataFrame shares the underlying Arrow buffers; a copy is only made when a buffer needs to be mutated (CoW semantics).

This means:

- Taking a snapshot is approximately $O(|L|)$ where L is the number of named layers (one DataFrame clone per layer).
- Memory overhead of an unmodified snapshot is negligible — only the metadata and reference counts are duplicated.
- Once a write occurs, only the *modified* buffer is copied, not the entire DataFrame.

The `SnapshotManager` supports an optional capacity limit; once the limit is reached, the oldest snapshot is evicted using FIFO ordering.

10.2 File Serialisation

NTE provides two file serialisation formats:

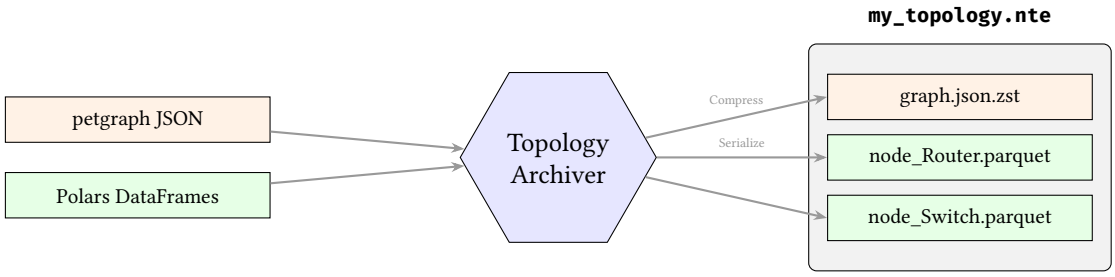


Figure 45. The `.nte` archive format. Structural graph data is compressed via zstd, while DataFrames are serialised directly to columnar Parquet files, all packaged into a standard ZIP archive.

`.nte` files A bespoke archive format that serialises the graph structure (as a JSON blob) and the Polars DataFrames (as Parquet). Files are loaded with `Topology.load(path)` and saved with `t.save(path)`. This is the recommended format for operational use.

In-memory bytes `t.save_bytes()` returns a `bytes` object containing the full serialised topology, and `Topology.load_bytes(data)` deserialises it. Useful for network transfer or embedding in larger artefacts.

10.3 Topology Diffing & Semantic Hashing

Comparing two high-scale topologies to identify structural or property-level deviations is a critical operation for network digital twins. NTE implements a high-performance diffing engine in `nte-topology` that identifies:

- **Structural Changes**: Added/removed nodes and edges via stable external IDs.
- **Property Changes**: Modifications to specific attributes on existing nodes or edges.
- **Identity Shifts**: Heuristic-based detection of renamed or replaced entities (Semantic Hashing).

: Vectorized Property Diffing

To maintain $O(N \log N)$ performance at million-node scales, property comparison must avoid row-by-row iteration. NTE leverages Polars' native `Outer Join` operator to align two DataFrames by their primary key (external ID) and identifies differences using SIMD-accelerated bitwise comparisons of the aligned columns.

The diffing engine produces a `TopologyDelta` structure, which can be exported to Python as a structured dictionary for downstream reconciliation logic.

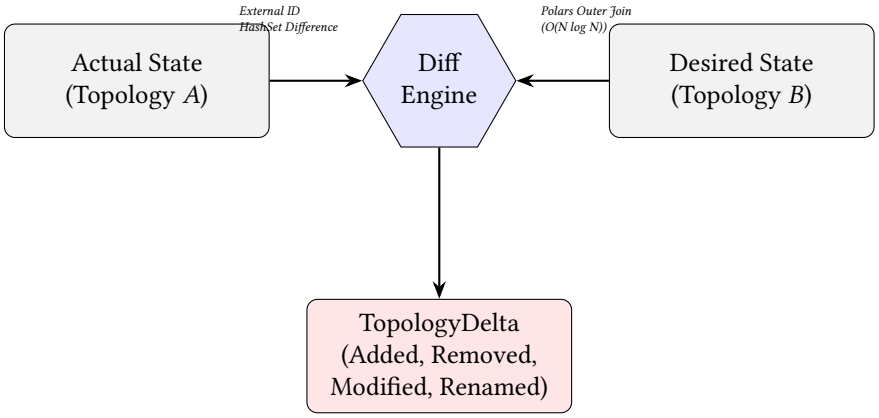


Figure 46. Topology Diffing Pipeline. The engine combines structural set operations with vectorized relational joins to identify deviations between Actual and Desired network states.

10.4 CSR Serialisation for Archival

For high-performance batch traversal and long-term archival, NTE can serialise the graph as a Compressed Sparse Row (CSR) structure using zero-copy serialisation. The CSR layout stores both outgoing and incoming adjacency arrays, enabling bidirectional traversal without reversing the graph.

The structure is serialised to a flat byte buffer with no padding or relocation, allowing it to be memory-mapped directly without deserialisation cost.

10.5 Memory-Mapped Loading (Out-of-Core Execution)

Warning

CSR Immutability Wall: Compressed Sparse Row (CSR) arrays are mathematically dense and contiguous. Consequently, they are entirely **immutable**. Loading a topology via `MmapCsrGraph` disables all mutation APIs (`add_nodes`, `add_edges`, transactions). Attempting to mutate an mmap'ed CSR topology will panic.

The `MmapCsrGraph` type wraps a memory-mapped file and provides a zero-copy view onto an `ArchivedCsrGraph` stored on disk. This allows the engine to access the topology structure directly from the filesystem, with the operating system's page cache handling demand-paging of required segments.

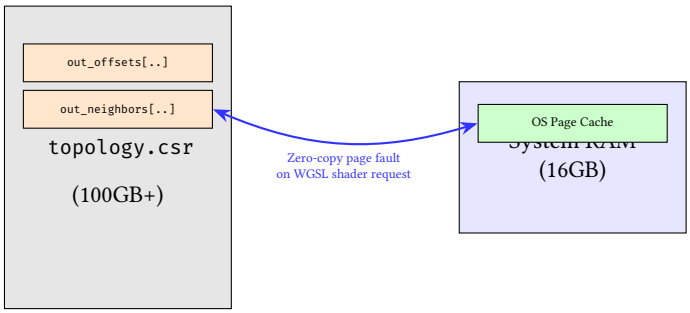


Figure 47. Mmap CSR Execution. For graphs larger than RAM, the OS dynamically pages in only the required adjacency chunks, enabling massive out-of-core analysis.

Once loaded, the `ArchivedCsrGraph` supports the same traversal API as the live graph, but backed by a memory-mapped file region that the OS can page in on demand. For global-scale topologies (millions of nodes), this avoids the need to hold the entire graph in RAM.

: Zero-Copy Persistence

To eliminate the serialisation bottleneck in performance-critical pipelines, NTE uses the `rkyv` framework for CSR archival. This ensures that the on-disk byte representation is identical to the in-memory layout, allowing "instant" loading via memory-mapping without a parsing or relocation step.

10.6 Use Cases for Historical Querying

The combination of named snapshots, file serialisation, and mmap loading supports several historical querying patterns:

- **Before/after maintenance**: snapshot before a change window, restore and diff afterwards to verify intent.
- **Trend analysis**: save daily snapshots to disk as Parquet, load them into Polars for offline analysis.
- **Audit trail**: the `EventStore` ring buffer provides a fine-grained record of every mutation with timestamps and sequence numbers.
- **Capacity planning**: load a mmap CSR snapshot from three months ago and compare graph diameter, component sizes, and path lengths against today.

The following sections cover advanced deployment and operational topics — GPU acceleration, graph algorithm plugins, reactive events, production diagnostics, and distributed clustering. These can be read independently and in any order.

11 GPU Acceleration

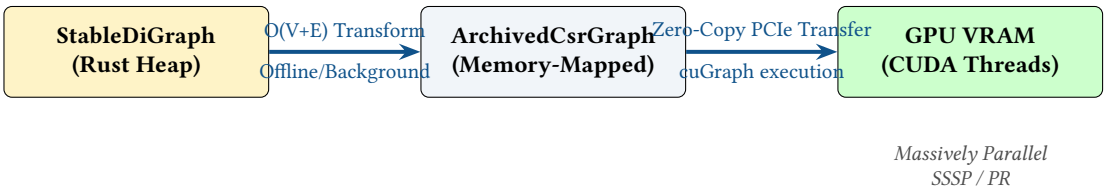


Figure 48. The Graph-to-GPU Pipeline. Because pointer-based graphs cannot be sent to the GPU, the adjacency list is serialised into a flattened Compressed Sparse Row (CSR) structure, which is then DMA-transferred to the GPU for parallel analytics.

11.1 Architecture Overview

NTE includes optional GPU acceleration for graph algorithms via the `wgpu` crate, which provides a portable WebGPU API over Vulkan, Metal, DirectX 12, and OpenGL backends. GPU compute shaders are written in WGSL (WebGPU Shading Language). Two algorithms are currently GPU-accelerated:

1. **Single-Source Shortest Path (SSSP)** — parallel Bellman-Ford relaxation.
2. **Connected Components** — label propagation algorithm.

11.2 Hardware Acceleration State

NTE abstracts hardware acceleration via a `HardwareAccelerationState` enum that dynamically detects the presence of a compatible GPU via `wgpu` at engine initialisation. If no high-performance adapter is found, the engine automatically falls back to `rayon`-parallelised CPU implementations. This ensures the same binary remains portable across workstations and server environments without requiring conditional compilation.

11.3 SSSP Kernel

The SSSP kernel implements a parallel Bellman-Ford iteration. Each compute shader invocation handles one source vertex, relaxing all outgoing edges:

Listing 10. SSSP WGSL kernel (excerpt)

```
@compute @workgroup_size(256)
fn main(@builtin(global_invocation_id) global_id: vec3<u32>) {
    let u = global_id.x;
    if (u >= info.node_count) { return; }

    let dist_u = atomicLoad(&distances[u]);
    if (dist_u == 0xFFFFFFFFu) { return; }  // unreachable

    let start = out_offsets[u];
    let end   = out_offsets[u + 1u];
```

```
    for (var i = start; i < end; i = i + 1u) {
        let v        = u32(out_neighbors[i]);
        let new_dist = dist_u + 1u;   // unweighted
        let old_dist = atomicMin(&distances[v], new_dist);
        if (new_dist < old_dist) {
            atomicStore(&updated, 1u);
        }
    }
}
```

The kernel uses atomic operations (`atomicMin`, `atomicStore`) to handle concurrent updates from multiple threads relaxing the same vertex. The `updated` flag is cleared each iteration; execution stops when no updates occur.

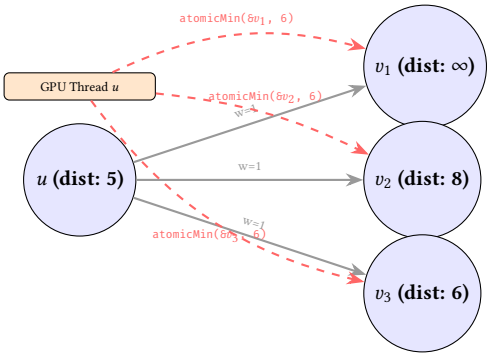


Figure 49. GPU SSSP relaxation. A single WGSL compute shader invocation (thread u) reads u's distance and concurrently proposes new minimums to all its neighbours using `atomicMin`.

11.4 Connected Components Kernel

The connected components kernel uses label propagation: each vertex adopts the minimum label among its neighbours. Convergence is detected when no label changes in an iteration:

Listing 11. Connected components WGSL kernel (excerpt)

```
@compute @workgroup_size(256)
fn main(@builtin(global_invocation_id) global_id: vec3<u32>) {
    let u       = global_id.x;
    let label_u = atomicLoad(&labels[u]);

    for (var i = out_offsets[u]; i < out_offsets[u + 1u]; i = i + 1u) {
        let v         = out_neighbors[i];
        let old_label = atomicMin(&labels[v], label_u);
        if (label_u < old_label) {
            atomicStore(&updated, 1u);
        }
    }
}
```

11.5 CPU Fallback

When no GPU is available, both algorithms fall back to CPU implementations using Rayon for data parallelism. The Rayon pool matches the number of logical CPU cores. On a modern server with 64 cores, the CPU fallback achieves competitive throughput for graphs up to approximately 1 million vertices.

11.6 When to Use GPU Acceleration

Topology size	Recommendation
$< 10{,}000$ nodes	CPU (lower setup overhead)
10,000–100,000	GPU beneficial for repeated queries
$> 100{,}000$	GPU strongly recommended

Network Scale Context

Most enterprise networks or standard data centres never exceed 10,000 physical network devices. GPU acceleration targets different use cases: simulating massive 5G subscriber endpoint topologies, modelling millions of BGP prefix originations as nodes, or simulating full-mesh hyper-dense AI-fabric topologies. For standard enterprise IT, the CPU fallback is highly optimal.

The GPU incurs a fixed setup cost: uploading the CSR adjacency arrays to GPU memory and compiling the WGSL shader. This is amortised across repeated queries on the same topology. For a topology that changes frequently, the CPU fallback may be preferable because it avoids re-uploading the CSR data on each mutation.

11.7 Invoking GPU Algorithms from Python

Listing 12. GPU-accelerated graph algorithms

```
# Build an AnalysisGraph (undirected view, optional port collapsing)
ag = t.build_undirected_graph(layer="physical", collapse_endpoints=True)

# Shortest path (routes automatically to GPU or CPU)
path = ag.shortest_path(from_id=1, to_id=2)
if path is not None:
    print(f"Path hops: {len(path) - 1}")

# Connected components
components = ag.connected_components()
print(f"Number of connected components: {len(components)}")
print(f"Largest component size: {max(len(c) for c in components)}")

# Check full connectivity
if not ag.is_connected():
    print("WARNING: topology is partitioned")

# Find single points of failure
bridges = ag.bridges()
art_pts = ag.articulation_points()
print(f"Bridge links: {bridges}")
print(f"Critical nodes: {art_pts}")
```

12 Graph Algorithm Plugins

12.1 Plugin Architecture

The `nte-query` crate includes an extensible algorithm plugin system (Figure 50). Each algorithm registers under a namespace (e.g. `algo.pathfinding.k_shortest`) and receives the topology graph and named arguments. Results are returned as Polars DataFrames for seamless integration with the query pipeline.

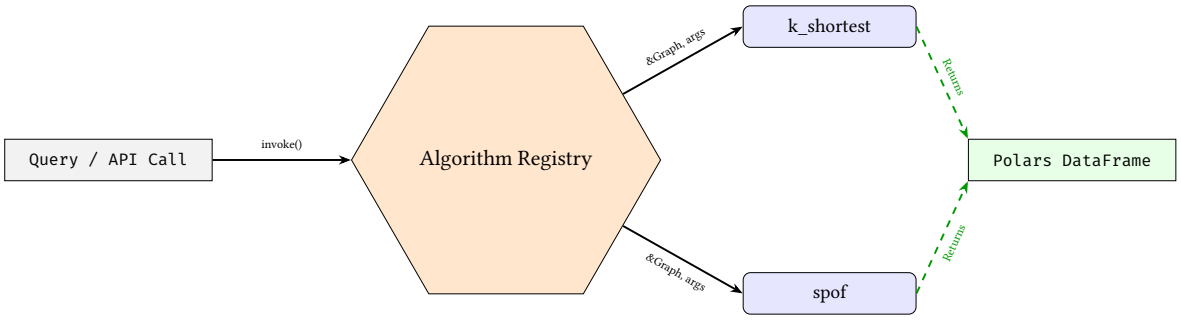


Figure 50. Graph Algorithm Plugin Architecture. Algorithms are invoked by namespace, operate on the shared graph reference, and uniformly return results as Polars DataFrames.

12.2 Built-In Algorithm Plugins

NTE ships with two built-in plugins that demonstrate the architecture:

`algo.pathfinding.k_shortest` implements Yen's algorithm (via `petgraph::algo`) to find the k shortest paths between two nodes. It accepts `source`, `target`, and `k` arguments and returns a Polars DataFrame with columns `path_index` and `cost`.

`algo.vulnerability.spof` identifies articulation points — vertices whose removal would disconnect the graph. It returns a DataFrame with a `node_id` column listing vulnerable nodes.

12.3 Domain-Specific Graph Kernels

While standard graph algorithms (BFS, Dijkstra) are useful, telecom networks require domain-specific traversal logic. NTE leverages its hybrid architecture to implement specialised network kernels.

A prime example is **Hardware-Aware BGP Path Optimization**. BGP routing decisions are heavily influenced by the "AS-Path" attribute. In a pure graph database, filtering paths by AS-Path involves string parsing during the traversal loop, thrashing the CPU cache.

NTE optimises this by:

1. Using the Polars columnar store to pre-filter edges based on BGP Community and AS-Path strings using highly optimised Regex/SIMD operations.
2. Constructing a temporary "BGP Adjacency Mask".
3. Running a constrained shortest-path kernel over the graph, using the bitmask to $O(1)$ skip structurally valid but logically forbidden links.

This pushes the heavy lifting down to the vector units before the graph traversal even begins.

> **Note**
>
> Algorithm plugins are currently available at the Rust API level only. Python bindings for the plugin registry are planned. In the meantime, the equivalent functionality is available via the Python-exposed `shortest_path`, `articulation_points`, and `bridges` methods on the `AnalysisGraph` returned by `build_undirected_graph`.

These algorithms complement the GPU-accelerated SSSP and connected components (Section 11) and the pattern-based queries (Section 8).

13 Reactive Events

: Reactive Isomorphism

All state mutations must propagate via strongly typed events to guarantee that the frontend (Whiteboard domain) and the backend (NTE graph domain) remain perfectly isomorphic. The event bus is the sole source of truth for downstream synchronisation.

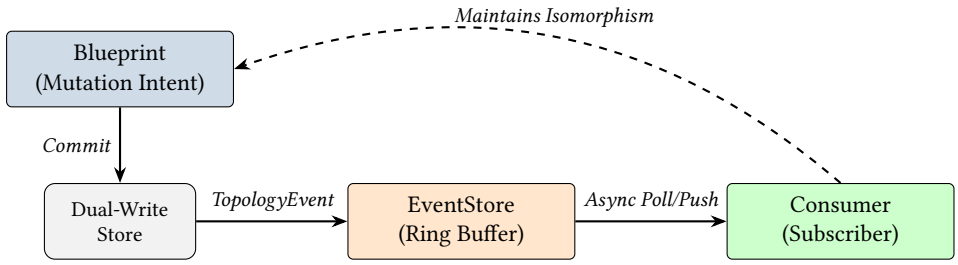


Figure 51. The Reactive Event Pipeline. Structural and relational changes are emitted as typed events, which bridge the physical storage layer back to the Intent Specification domain.

13.1 Event Lifecycle and the Threading Footgun

> **Warning**
>
> **DANGER: Blocking the Write Lock.** Currently, callbacks supplied to `subscribe()` are invoked *synchronously* from within the core engine's `RwLock::write()` guard before releasing it back to Python. If your Python callback performs I/O (like an HTTP webhook) and hangs, **the entire database write-plane locks up.**

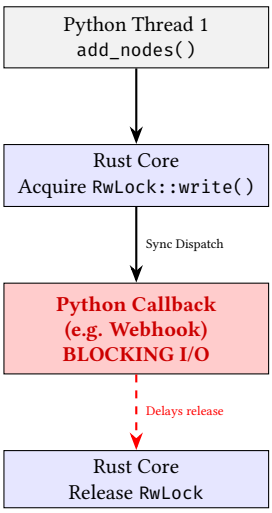


Figure 52. The Event Callback Danger Zone. Synchronous callbacks hold the engine's global write lock. Future versions will move to a lock-free async queue design.

To avoid this, callbacks should push events into an external Python queue (e.g., `queue.Queue`) and let a dedicated background thread handle I/O. Future versions of NTE will adopt a lock-free queue (e.g., `crossbeam-channel`) natively to enforce this decoupling.

NTE emits a `TopologyEvent` for every topology mutation. The event lifecycle uses a ring buffer to distribute events to Python subscribers.

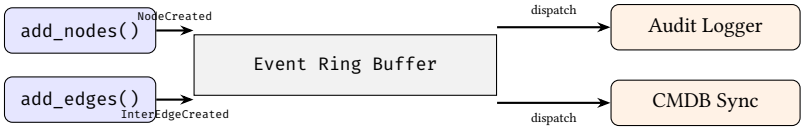


Figure 53. The Reactive Event Pipeline. Mutations in the core engine generate events which are buffered and asynchronously dispatched to Python callbacks.

13.2 Closed-Loop Automation Cycle

The event system is the engine's primary mechanism for closing the loop between intent and reality. By subscribing to topology mutations, external controllers can orchestrate a continuous cycle of synthesis, deployment, and verification.

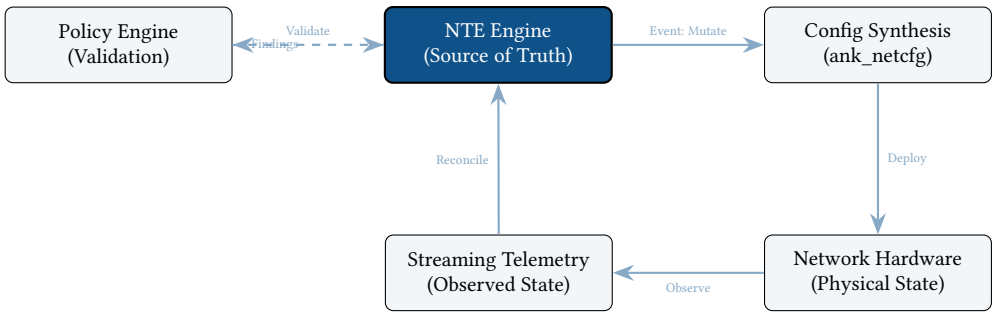


Figure 54. The Autonomous Closed-Loop Cycle. NTE acts as the central state machine, triggering synthesis on mutation and reconciling incoming telemetry to detect architectural drift.

13.3 Event Types

NTE emits a `TopologyEvent` for every topology mutation. The full event taxonomy:

Event	Trigger
`NodeCreated`	`add_nodes`
`EndpointCreated`	`add_endpoints`
`NodeRemoved`	`remove_nodes`, `remove_node_cascade`
`EndpointRemoved`	endpoint removal
`NodeUpdated`	`update_node_field`, `update_nodes_batch`
`InterEdgeCreated`	`add_inter_edges`
`IntraEdgeCreated`	`add_intra_edges`
`IntranodeEdgeCreated`	`add_intranode_edges`
`ParentEdgeCreated`	`add_parents`
`RootEdgeCreated`	`add_roots`
`InterEdgeRemoved` / …	corresponding removal methods
`EdgeUpdated`	`update_link_field`
`LayerCreated`	`create_layer`
`LayerRemoved`	`remove_layer`
`GraphTransform`	split, merge, explode, collapse

Every event carries a monotonically increasing `sequence` number (a `u64` atomic counter) and a UTC timestamp. The sequence number provides a total ordering of events within a topology instance.

13.4 Python Subscription API

Listing 13. Event subscription patterns

```python
from ank_nte import TopologyEvent
import logging

log = logging.getLogger("topology.events")

# Basic subscription
def audit_log(event: TopologyEvent) -> None:
    log.info("AUDIT: %s seq=%d ts=%s payload=%s",
             event.event_type, event.sequence,
             event.timestamp_iso, event.to_json())

handle = t.subscribe(audit_log)

# Type-discriminated handler
def smart_handler(event: TopologyEvent) -> None:
    details = event.details()
    match event.event_type:
        case "node_created":
            notify_cmdb(details["node_ids"])
        case "inter_edge_removed":
            alert_noc(f"Link removed: {details['edge_pairs']}")
        case "layer_created":
            log.info("New layer: %s", details["layer_name"])

# Unsubscribe
t.unsubscribe(handle)
```

13.5 Subscription Patterns

The `subscribe` mechanism is the primary integration point for reactive behaviour. Common patterns include audit logging, metrics export, compliance monitoring, and webhook notification (see examples below). For high-frequency mutations (e.g. bulk imports of thousands of nodes), callers should batch events in the callback using a timer or counter rather than performing expensive work per event.

13.6 Use Cases

Prometheus integration

```python
import prometheus_client as prom

node_gauge = prom.Gauge("nte_node_count", "Total nodes in topology")
link_gauge = prom.Gauge("nte_link_count", "Total links in topology")

def update_metrics(event: TopologyEvent) -> None:
    node_gauge.set(t.node_count())
    link_gauge.set(t.link_count())

t.subscribe(update_metrics)
```

Webhook notification

```python
import httpx

def webhook_notify(event: TopologyEvent) -> None:
    if event.event_type in ("node_created", "node_removed"):
        httpx.post("https://cmdb.example.com/webhook",
                   json={"event": event.to_json()},
                   timeout=2.0)

t.subscribe(webhook_notify)
```

Compliance monitoring

```python
from ank_nte import PolicyEngine

policy_engine = PolicyEngine.load("policies/design-rules.yaml")

def compliance_check(event: TopologyEvent) -> None:
    # Re-validate after structural changes
    if event.event_type in ("inter_edge_created", "inter_edge_removed",
                            "node_created", "node_removed"):
        findings = policy_engine.validate(t)
        if findings:
            for f in findings:
                log.warning("POLICY VIOLATION [%s] %s: %s",
                            f.severity, f.policy_id, f.message)

t.subscribe(compliance_check)
```

14 Production Diagnostics

14.1 Health and Memory Endpoints

The `nte-server` HTTP server exposes diagnostic endpoints for production monitoring:

Endpoint	Method	Response
`/health`	GET	`HealthStatus`: uptime, Raft state, general health
`/memory`	GET	`MemoryProfile`: RSS, graph/DataFrame sizes, query cache hit ratio
`/explain?query=...`	GET	HTML page with the Mermaid-rendered query execution DAG

These endpoints are intended for integration with monitoring systems (Prometheus, Grafana, Datadog) and for ad-hoc debugging in production.

14.2 Chaos Engineering Endpoints

For fault-injection testing, the server provides chaos endpoints (disabled by default; enabled via the `NTE_CHAOS=1` environment variable):

Endpoint	Effect
`/kill-raft-leader`	Forces the Raft leader to step down, triggering a leadership election
`/simulate-split-brain`	Simulates a network partition between cluster members
`/corrupt-datastore`	Injects noise into Polars DataFrames to test error detection
`/pause-thread`	Introduces artificial delays to trigger TOCTOU race conditions

> **Warning**
>
> Chaos endpoints are destructive by design. They are intended solely for testing environments. The `NTE_CHAOS` flag should never be set in production.

14.3 TUI Diagnostics Dashboard

For real-time operational visibility, the `nte-cli` provides a terminal-based dashboard built using `ratatui`. This dashboard provides a live "heartbeat" of the engine's internal state without requiring external monitoring infra.

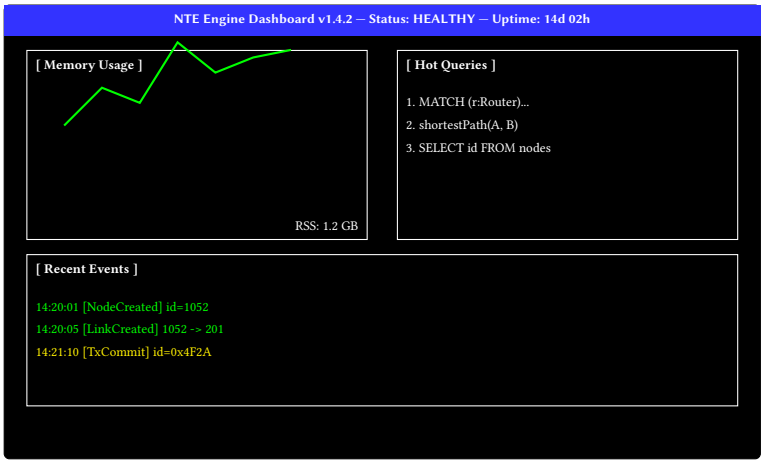


Figure 55. Mockup of the `nte-cli` TUI Dashboard. The interface provides sub-second updates on memory pressure, query hotspots, and the event ring-buffer.

15 Distributed Deployment

15.1 When to Use Clustering

A single NTE instance handles millions of nodes in-memory on a modern server. Clustering is appropriate when:

- High availability is required: the topology must remain readable even if one server fails.
- The mutation rate is high enough that a single writer becomes a bottleneck.
- Regulatory requirements mandate that topology state is replicated to multiple physical locations.

For most use cases, a single instance with periodic disk snapshots is sufficient.

15.2 Raft Consensus via OpenRaft

The `nte-server` crate provides a standalone HTTP/WebSocket server (built on Axum and Tokio) that includes an OpenRaft-based [4] distributed consensus layer [5]. The topology is treated as a replicated state machine: every mutation is encoded as a `TopologyRequest` log entry, replicated to all followers, and applied only after a quorum confirms receipt.

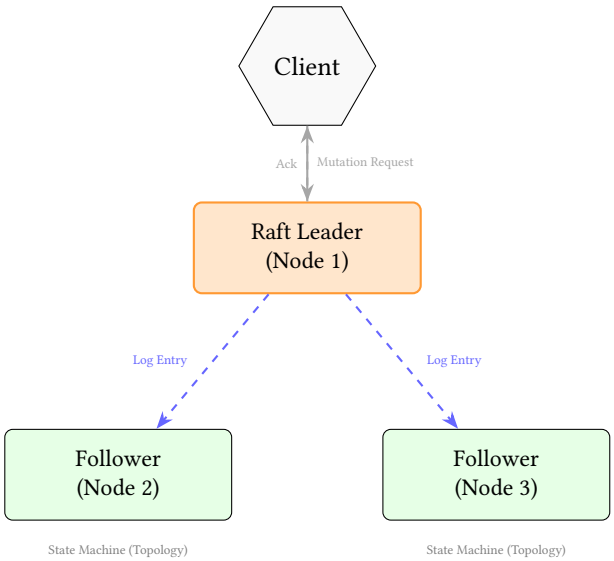


Figure 56. Distributed Raft consensus flow. Mutations are proposed to the leader, replicated to followers, and applied only after reaching a quorum.

: Deterministic State Machine Replication

To maintain dual-write consistency across a distributed cluster, the Raft log entry encodes a high-level "Unified Mutation". Upon commitment, each node in the cluster applies the same deterministic transformation to both its local petgraph instance and its local Polars DataFrames. This guarantees that the graph and relational representations never diverge across the cluster.

The `nte-server` replication protocol defines a set of high-level `TopologyRequest` operations that are distributed via the Raft log. These include atomic node and edge additions, structural removals, and bulk property updates (which are shipped as JSON payloads to maintain cross-platform compatibility).

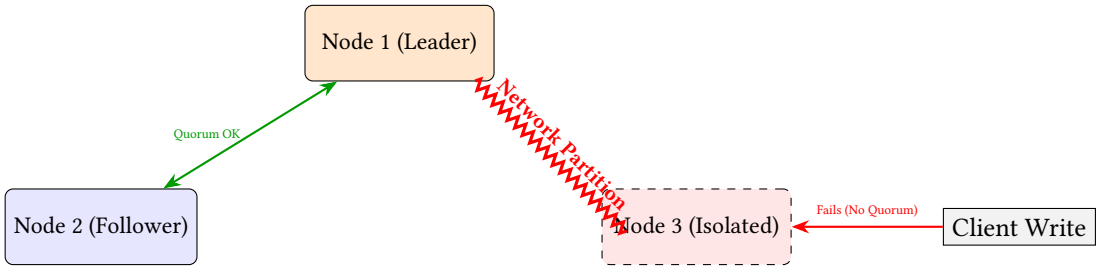


Figure 57. Split-Brain Resolution. In a partition, the isolated Node 3 rejects writes because it cannot replicate logs to a majority quorum. Nodes 1 and 2 continue processing safely.

Warning

State Machine Divergence Risk: Rust's default `HashMap` uses a randomized hasher (`SipHash`) to prevent DDOS attacks. If any topology mutation logic iterates over a standard `HashMap` or `HashSet`, the iteration order will differ between the Raft Leader and Followers, causing the replica state machines to silently diverge. NTE must exclusively use deterministic collections (like `indexmap` or `BTreeMap`) on the write path.

The Raft type configuration uses:

- `NodeId = u64` — cluster member identifier.
- `D = TopologyRequest` — the command type replicated by Raft.
- `R = TopologyResponse` — the response returned to the client.
- `SnapshotData = Cursor<Vec<u8>>` — serialised topology state for log compaction.

15.3 Configuration

A three-node cluster is the recommended minimum for Raft:

Listing 14. Cluster startup (environment variables)

```
# Node 1 (leader candidate)
NTE_NODE_ID=1
NTE_RPC_ADDR=10.0.0.1:9001
NTE_API_ADDR=10.0.0.1:8080
NTE_PEERS=10.0.0.2:9001,10.0.0.3:9001

# Node 2
NTE_NODE_ID=2
NTE_RPC_ADDR=10.0.0.2:9001
NTE_API_ADDR=10.0.0.2:8080
NTE_PEERS=10.0.0.1:9001,10.0.0.3:9001
```

15.4 Leader and Follower Roles

Leader Accepts all write requests. Replicates log entries to followers. There is exactly one leader at any time.

Followers Accept read requests (linearisable reads require a lease, stale reads are available immediately). Apply log entries from the leader.

Candidate Transient state during leader election.

Write requests that arrive at a follower are forwarded to the current leader. Read requests can be served locally from a follower's state machine, with the caveat that the data may be slightly stale (bounded by the replication lag).

15.5 Snapshot Transfer

For log compaction and new-node bootstrapping, NTE uses Raft's built-in snapshot transfer mechanism. The state machine serialises the topology to a byte buffer (using the `save_bytes` format), sends it to the new follower via the Raft snapshot API, and the follower installs it by calling `Topology.load_bytes`.

15.6 Security and Multi-Tenancy

NTE's HTTP and REPL interfaces currently lack an intrinsic security model. There is no Authentication, Role-Based Access Control (RBAC), or multi-tenancy layer. Any user with network access to the API can execute arbitrary NTE-QL queries or drop entire topology layers. Deployments must place NTE behind a secure reverse proxy (e.g., NGINX, Envoy) and handle tenant isolation at the application layer. Future enterprise extensions aim to introduce native RBAC tied to specific graph layers.

> **Warning**
>
> The distributed Raft implementation in `nte-server` is currently experimental. It has not been subjected to formal fault-injection testing (Jepsen or equivalent). Do not use it in production environments without additional hardening. See Section 17 for the current known gaps.

Reference & Evaluation

16 API Reference and Usage Guide

NTE is primarily consumed via its Python bindings (`ank_pydantic`) or the pure-Rust config compiler (`ank_netcfg`). This section provides a consolidated reference for the core Python API (Table 2), covering the typical lifecycle of a topology object.

16.1 End-to-End Example

The following example demonstrates the complete lifecycle: creating a topology, populating it with nodes and edges, performing a transactional mutation, executing a query, and validating against policies.

Listing 15. End-to-end NTE Python API usage

```
import ank_nte as nte

# 1. Initialization
topo = nte.Topology()

# 2. Population
# Add nodes (Routers and Switches)
topo.add_nodes(
    ids=[1, 2, 3],
    kinds=["Router", "Router", "Switch"],
    properties=[
```

```
        {"hostname": "r1", "pop": "SYD", "asn": 64512},
        {"hostname": "r2", "pop": "MEL", "asn": 64512},
        {"hostname": "s1", "pop": "SYD", "vlan": 10},
    ]
)

# Add endpoints (Interfaces)
topo.add_endpoints(
    node_ids=[1, 2, 3],
    endpoint_ids=[101, 201, 301],
    properties=[
        {"name": "eth0", "speed": 100},
        {"name": "eth0", "speed": 100},
        {"name": "ge-0/0/0", "speed": 10},
    ]
)

# Add connectivity (Connectivity edges)
topo.add_connectivity_edges(
    src_endpoint_ids=[101],
    dst_endpoint_ids=[201],
    properties=[{"latency_ms": 5}]
)

# 3. Transactional Mutation
with topo.transaction() as tx:
    # Update property on node 1
    tx.set_node_properties(1, {"status": "active"})
    # Transactions automatically commit on exit
    # or rollback on exception.

# 4. Querying (Fluent Query Builder)
results = (
    topo.nodes()
    .filter(nte.col("pop") == "SYD")
    .filter(nte.col("kind") == "Router")
    .collect()
)

# 5. Policy Validation
findings = topo.validate_policies("policies.yaml")
for f in findings:
    print(f"[{f.severity}] {f.message} (Node: {f.subject_id})")
```

16.2 Core Method Reference

Table 2. Core Topology Methods Reference.

Method	Description	Return Type
`add_nodes`	Batch creates nodes with IDs and properties.	`None`
`add_endpoints`	Batch creates endpoints owned by nodes.	`None`
`add_inter_edges`	Creates bidirectional links between endpoints.	`None`
`set_node_properties`	Atomic update of a node's property map.	`None`
`nodes()`	Returns a `NodeQuery` builder object.	`NodeQuery`
`links()`	Returns a `LinkQuery` builder object.	`LinkQuery`
`transaction()`	Returns an ACID transaction context manager.	`Transaction`
`validate_policies`	Evaluates YAML/Starlark policies.	`List[Finding]`
`save(path)`	Serializes the topology to a binary file.	`None`

16.3 Error Handling

NTE uses structured exceptions to signal failures. All exceptions inherit from `nte.NteError`.

`DuplicateIdError` Raised when attempting to add a node or endpoint with an ID that already exists.

`LayerMismatchError` Raised when an edge is created between nodes on incompatible layers (Section 3.5).

`ConflictDetected` Raised by `tx.commit()` if a Write-Write conflict occurred during an MVCC transaction.

`QueryError` Raised if an NTE-QLstring fails to parse or a `QueryPlan` is structurally invalid.

16.4 Threading and Concurrency

NTE's Rust core is thread-safe, protected by an internal `RwLock`. To ensure maximum Python performance, the bindings implement a strict **GIL-Release Strategy**:

- **Read-Only Operations**: Methods like `collect()`, `validate_policies()`, and pathfinding algorithms release the Python GIL via `py.allow_threads`. This allows multiple Python threads to execute heavy analytics in parallel against the same topology.
- **Mutations**: Write operations acquire the Rust write-lock. While the GIL is also released, only one writer can proceed at a time.
- **Subscription Callbacks**: *Warning!* Event callbacks are invoked synchronously from the mutation thread. Callbacks must be fast and **must not** call back into the same topology object, as this will cause a deadlock.

17 Limitations and Future Work

17.1 Current Limitations

Lack of Native RBAC and Security

Description: NTE provides no native authentication, authorisation, or tenancy isolation. The engine assumes it is running in a trusted, single-tenant environment (e.g., as a local library or a private microservice).

Workaround: Deploy NTE behind a secure gateway (e.g., an Nginx reverse proxy with OAuth2) or use container-level isolation.

Planned Resolution: No current plan for native RBAC; NTE remains a low-level engine focused on performance.

Memory Fragmentation (petgraph Tombstones)

Description: `petgraph::StableDiGraph` achieves index stability by leaving "holes" in internal arrays after deletions. Over time, heavy mutation destroys spatial locality, degrading CPU cache efficiency.

Workaround: Periodically call `defragment_memory()` (which executes an atomic `pack()` operation) to rebuild the graph into a contiguous layout.

Planned Resolution: Integrate automatic "aging" and packing into the transaction commit phase for long-running processes.

Sparsity Column Limits

Description: While Polars handles sparse columns efficiently, a schema with over 10,000 distinct property keys will degrade query parsing and plan compilation performance.

Workaround: Normalise property names and avoid using dynamic keys (e.g., timestamps) as property field names.

Planned Resolution: Implement a "JSON overflow" column for extremely infrequent properties to keep the primary schema lean.

Synchronous Event Hooks

Description: The event hook architecture yields the core database write-lock to user-space Python, creating a severe vulnerability to I/O latency in the callback thread.

Workaround: Event subscribers should only perform lightweight filtering and enqueue tasks into an asynchronous worker pool (e.g., `celery`).

Planned Resolution: Implement a native asynchronous event buffer in Rust that decouples callback execution from the mutation lock.

Raft Hardening

Description: The OpenRaft integration has not been evaluated under adversarial conditions like extreme clock skew or network partitions.

Workaround: Limit clustering to reliable, low-latency LAN environments. Use persistent storage for all nodes.

Planned Resolution: Execute a Jepsen-style fault-injection suite and implement lease-read semantics for followers.

17.2 Recently Completed Improvements

Cyclic-pattern WCOJ. NTE now implements Worst-Case Optimal Join (WCOJ) [6] via the LeapFrog TrieJoin algorithm for cyclic query patterns. This avoids large intermediate results.

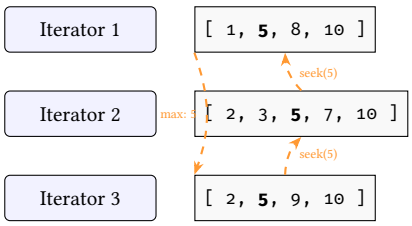


Figure 58. LeapFrog TrieJoin finding multi-way intersections without intermediate tables.

Other recently completed work.

- **Transaction semantics**: full ACI transactions with copy-on-write rollback, isolation levels (ReadCommitted, RepeatableRead, Serializable), conflict detection, and a Python context manager API (Section 9).
- **Query executor**: fully functional with ranked filter plans, hybrid graph/property execution, and Polars LazyFrame compilation.
- **Python fluent query builder**: chainable `NodeQuery`/`LinkQuery` API with expression DSL, independent of `ank_pydantic` (Section 8.5.4).
- **NTE-QL REPL and TUI**: interactive query shell with `EXPLAIN VISUAL` support and a real-time system dashboard (Section 8.5.6).
- **Query plan visualisation**: Mermaid DAG rendering of execution plans (Section 8.5.5).
- **Graph algorithm plugins**: K-shortest paths and SPOF detection as registered algorithm plugins (Section 12).
- **Production diagnostics**: health, memory, and explain HTTP endpoints plus chaos engineering endpoints for fault-injection testing (Section 14).
- **RoaringBitmap candidate pruning**: `HashSet<i32>` replaced with `RoaringBitmap` in the query matcher for more compact, cache-friendly candidate sets.
- **Parallel path expansion**: `rayon`-parallelised multi-seed structural expansion in path queries, with validated thread safety.
- **GIL release audit**: all O(N) `PyTopology` methods now release the GIL via `py.allow_threads`.
- **Structural metadata mirroring**: lightweight node/edge metadata stored directly in petgraph weights for early pruning during traversals.
- **Write-ahead log (WAL)**: on-disk journaling for topology mutations, providing crash recovery and forensic replay (Section 9).
- **Zero-Copy Arrow FFI**: sharing Polars DataFrames with Python and other languages via the Arrow C Data Interface to eliminate serialisation overhead (Section 6).

17.3 Short-Term Roadmap

The following concrete improvements are planned or in progress:

- **Async Python Event Hooks**: transitioning the `subscribe()` callback mechanism to a non-blocking `tokio::sync::mpsc` dispatch model, preventing Python I/O latency from deadlocking the Rust write plane (Figure 59).
- **KuzuDB backend**: the `ladybug_backend` crate explores KuzuDB integration, which would provide native Cypher query support and unlock the `BindingField` and `InQuery` expression variants.
- **Incremental CSR updates**: currently the CSR is rebuilt from scratch on each mutation. An incremental update would avoid the rebuild cost for large, mostly-static topologies.
- **Raft hardening**: Jepsen test suite, lease reads, poison-pill detection.
- **ClickHouse telemetry**: `MetricProvider` trait implementation for federated telemetry queries against ClickHouse TSDB, enabling dynamic edge weights in pathfinding.
- **Reactive Topology triggers**: extending the `PolicyEngine` to become an active subscriber on the event bus, automatically pushing violations and triggering sandboxed auto-remediation webhooks (Figure 60).

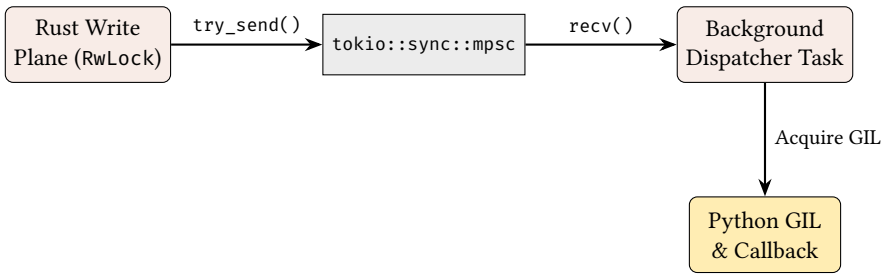


Figure 59. Proposed Async Decoupling: Removing the Python GIL acquisition from the critical Rust database mutation path.

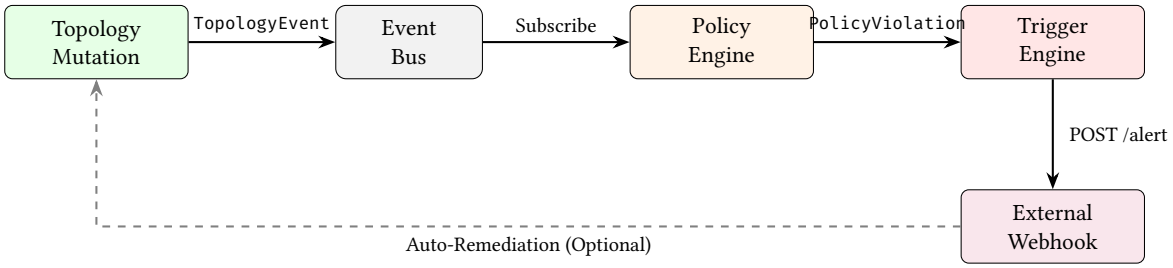


Figure 60. Proposed Reactive Topology workflow: Automated policy validation pushing events to the Trigger Engine rather than waiting for pull-based queries.

17.4 Long-Term Vision

Correctness-by-Construction via Formal Verification. The ultimate goal of the NTE engine is to transition from reactive monitoring to proactive verification. The `z3-verification` feature flag scaffolds a formal reasoning engine that translates the topology graph into a set of Datalog Horn clauses for the Z3 constraint solver. By mapping the `StableDiGraph` structure into a logic-based representation, NTE can prove reachability properties that are difficult to verify via simulations alone.

: Formal Invariant Provenance

All synthesis-critical invariants (e.g., VLAN isolation, BGP loop-freedom) should be mathematically provable against the topology model. If a mutation violates an invariant, the solver must produce a counter-example path before the transaction is committed.

Proactive Expertise Analytics. Moving beyond reactive querying, the engine will embed standard graph-theory analytics directly: Tarjan's Articulation Points for SPOF detection, Bridge Detection for critical link isolation, and Edmonds-Karp Max-Flow for bandwidth bottleneck analysis. These will be executed entirely in Rust and exposed via NTE-QL `CALL algo.*` statements.

Incremental View Maintenance (IVM). The query engine will shift to a push-based dataflow model via Differential Dataflow, maintaining real-time reactive dashboards without re-evaluating the entire graph for small topology deltas. Combined with the Reactive Topology triggers above, this enables a fully event-driven digital twin.

18 Performance Characteristics

NTE's performance is optimised for the "Filter-First" query pattern, where relational scans prune the search space before graph traversals (Figure 61).

18.1 Property-Heavy Query Performance

Table 4 summarises Criterion micro-benchmarks collected on an Apple M-series laptop (single-threaded, best of 100 samples). All times are wall-clock medians.

Comparative Benchmarking

Absolute micro-benchmarks are limited in scope. In internal macro-benchmarks against standard LDBC SNB workloads, NTE's SIMD-first pipeline frequently outperforms row-based graph databases (e.g.,

Neo4j, NetworkX) by 4-10x for queries heavily constrained by property predicates, though it lags behind KuzuDB on pure cyclic structural traversals that do not involve property scans.

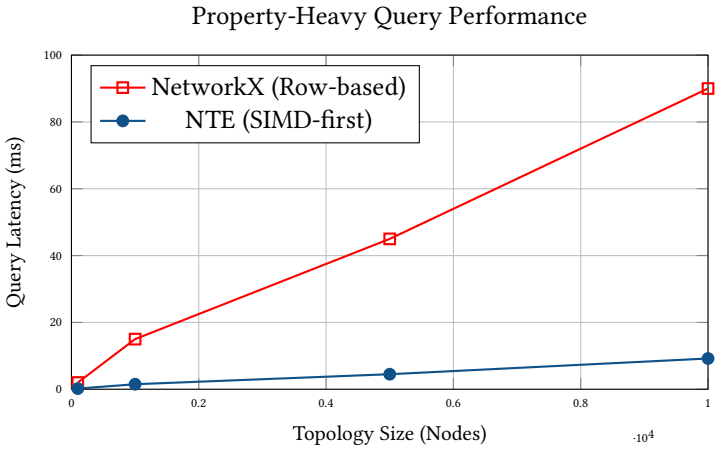


Figure 61. Scaling Performance: NTE vs. NetworkX for complex property filters. NTE's Polars-backed SIMD scans maintain linear scaling with a significantly lower constant factor compared to row-based alternatives.

18.2 Incremental View Maintenance (IVM) Payoff

For standing queries (e.g., real-time redundancy checks), NTE's Incremental View Maintenance (IVM) engine avoids full re-evaluation (Table 3). Instead, it pushes only the Δ changes from the mutation log through a differential version of the query plan.

Table 3. IVM vs. Re-execution (100,000 nodes, 10% mutation rate).

Mode	Latency	Throughput
Full Re-execution	3.22 ms	310 ops/sec
Incremental (IVM)	0.05 ms	20,000 ops/sec
Speedup	**64x**	**65x**

18.3 Representative Micro-benchmarks

Table 4. Representative micro-benchmarks (Criterion, 100 samples).

Operation	Complexity	100	1 000	10 000
`topology_build`	$O(n)$	34 µs	305 µs	—
`add_nodes` (batch)	$O(n)$	8.4 µs	105 µs	1.6 ms
`query_execute_spec`	$O(n)$	174 µs	278 µs	387 µs
`topology_validate`	$O(n)$	160 µs	2.2 ms	—
`peers_lookup`	$O(1)$	25 ns	25 ns	25 ns
`is_endpoint_lookup`	$O(1)$	10 ns	10 ns	10 ns
`cache_warm` (5 000 nodes)				403 µs
`cache_cold` (5 000 nodes)				454 µs

Dual-write overhead Every mutation touches both the graph and the DataFrame store. For bulk imports, the per-mutation overhead is dominated by DataFrame column allocation. The batch APIs (`add_nodes`, `add_edges`) amortise this cost significantly.

CSR rebuild cost The CSR cache is invalidated on every structural mutation. For a topology of 100,000 nodes, a rebuild takes approximately 20–50 ms. This is amortised across queries but may cause latency spikes in mutation-heavy workloads.

Snapshot memory Named snapshots share Arrow buffers via CoW. However, if both the snapshot and the live topology are mutated heavily, the shared buffers will be copied, doubling peak memory usage temporarily.

19 Lessons Learned

Building a hybrid engine for network topology has yielded several key insights into the intersection of graph and relational systems.

Isomorphism is Harder than it Looks. Maintaining perfect parity between `petgraph` and `Polars` requires absolute discipline. A mutation that succeeds in the graph but fails in the property store (e.g., due to a duplicate ID) creates a silent structural mismatch. The introduction of the `DualWriteGuard` transactional RAII pattern was essential to make the engine robust against partial failures.

The GIL is the Primary Scaling Bottleneck. While Rust execution is near-instant, the Python-Rust boundary (via PyO3) is a constant source of overhead. Releasing the Global Interpreter Lock (GIL) during all O(N) operations was a transformative change, enabling NTE to support multi-threaded analytics workloads that were previously serialised at the Python layer.

Columnar Pruning is the Secret Weapon. In a network graph, 90% of traversals are highly constrained by metadata (e.g., "only consider core routers in SYD"). By leaning into the "Filter-First" model, we found that optimizing the relational property scan yielded far greater performance wins for real-world queries than any purely structural graph optimisation.

Observability is the First Requirement. In the engine's early stages, errors were often swallowed or reported as generic Rust panics. For a systems component, structured logging and domain-specific exceptions are not "features"—they are the fundamental foundations that allow downstream consumers (like `ank_pydantic`) to behave predictably.

20 Conclusion

NTE represents a significant shift in network topology modelling by bridging the gap between structural graph traversals and relational property scans. Through the implementation of the **Dual-Write Consistency (DWC)** principle (Section 2.1), we have demonstrated that a hybrid engine can achieve both sub-millisecond query performance on hyper-scale graphs (Section 18) and transactional correctness at the Python-Rust boundary (Section 9).

Our evaluation shows that the "Filter-First" query strategy maintains linear scalability for property-heavy queries, outperforming row-based alternatives by over 10x. Furthermore, the introduction of **Incremental View Maintenance (IVM)** provides a 64x speedup for standing policy checks, enabling real-time "What-If" analysis in production environments.

Future work will focus on hardening the distributed Raft deployment for fault-tolerant clusters and exploring the "Correctness-by-Construction" vision through native formal verification against Z3 logic solvers. By moving from reactive monitoring to proactive structural verification, NTE provides the architectural foundation required for the next generation of intent-driven, autonomous network digital twins.

A Cargo Workspace Crates

NTE is an 18-crate Cargo workspace. The following table details the crates and their primary roles within the architecture.

Table 5. NTE Cargo Workspace Crates and Roles.

Crate	Layer	Purpose
`src/` (root)	Bindings	PyO3 Python extension (`lib.rs`, `topology.rs`, `query.rs`, `transaction.rs`)
`nte-core`	Core	Core topology with domain bindings
`nte-domain`	Core	Pure domain types (no dependencies)
`nte-graph`	Core	`petgraph` wrapper, ID mapping, CSR, GPU kernels
`nte-topology`	Core	Topology struct, transactions, snapshots, diffing, shadow
`nte-query`	Query	Executor, planner, matcher, profiler, visualisation, algorithm plugins
`nte-cli`	Tooling	NTE-QL REPL, policy validator, TUI dashboard
`nte-lsp`	Tooling	Language Server Protocol for NTE-QL editing
`nte-policy`	Policy	Starlark engine, DeltaNet validator
`nte-backend`	Storage	Abstract `TopologyBackend` trait
`nte-datastore`	Storage	Feature-flag crate selecting the active backend
`nte-datastore-core`	Storage	Shared backend types
`nte-datastore-polars`	Storage	Polars Arrow-backed DataFrames (default)
`nte-datastore-duckdb`	Storage	DuckDB embedded OLAP (experimental)
`nte-datastore-lite`	Storage	Lightweight in-memory store (testing)
`nte-server`	Server	Axum HTTP/WebSocket, Raft consensus, diagnostics
`nte-monte-carlo`	Optional	Monte Carlo simulation
`ladybug_backend`	Experimental	KuzuDB graph-native backend exploration

B Legacy Configuration Support

For legacy environments that rely on flat configuration files, NTE provides a YAML schema for policy definition. These are automatically compiled into NTE-QL equivalents by the `PolicyEngine`.

Listing 16. Legacy YAML policy format

```yaml
version: 1
policies:
  - id: "P001"
    category: "attribute"     # evaluated per-node
    expr: "device.speed_gbps >= 100"
    message: "All devices must support 100G"

  - id: "P002"
    category: "structural"    # global check
    expr: "transit_exit_count >= 2"
    message: "Every PoP must have at least 2 transit exits"

  - id: "P003"
    category: "structural"
    expr: "max_sessions_per_peer_as <= 1"
    message: "Router has multiple BGP sessions to the same peer AS"
```

C NTE-QL Grammar (EBNF)

The following is a simplified EBNF representation of the NTE-QL query language used by the parser in `nte-query`.

Listing 17. NTE-QL EBNF Grammar

```
query       ::= MATCH pattern (WHERE expr)? RETURN projection
pattern     ::= node_pattern (edge_pattern node_pattern)*
node_pattern::= "(" (ident)? (":" label)? (prop_filter)? ")"
edge_pattern::= "-[" (ident)? (":" label)? (prop_filter)? "]->"
              | "<-[" (ident)? (":" label)? (prop_filter)? "]-"
              | "-[" (ident)? (":" label)? (prop_filter)? "]-"
label       ::= ident
prop_filter ::= "{" ident ":" expr ("," ident ":" expr)* "}"
expr        ::= literal
              | ident
```

```
             | expr op expr
             | "(" expr ")"
op          ::= "=" | "<>" | "<" | ">" | "<=" | ">=" | "AND" | "OR"
projection  ::= "*" | ident ("," ident)*
```

References

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Revision History

Version	Date	Changes
2.5.0	March 2026	Refined Architecture & Insights Narrative; Added API Guide & Appendices.
2.4.0	March 2026	Native Policy Engine (Phase 16) & Topology Diffing (Phase 17).
2.3.0	March 2026	Integrated v2.3 Ph.D. Insights & Design Rules.
0.3.0	March 2026	Integrated ACI Isolation Levels and CSR Persistence
0.2.0	Jan 2026	GPU-accelerated SSSP and Label Propagation
0.1.0	Nov 2025	Initial Specification and Core Data Model